ENGINEER

Zeeshan Mumtaz

Industrial Engineer | Design Engineer



Mechanical systems, CAD assemblies and practical engineering.

17 engineering and academic projects, including vehicle platforms, cooling systems, fabrication work and lean healthcare process improvement.

qazizeeshan350@gmail.com | +92 341 3642355

PROJECT INDEX

A record of engineering work.

01 Barrel Cleaning Robot - 125 mm Tank Barrel

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13 Aviation Bowser

14 AC Truck - 16-Ton Cooling System

15 Dragon AC Cooling Cart

16 Academic Layout Design & Equipment Placement

17 Analysis Of Waste And Designing Of Lean Systems for Healthcare

The project sequence ends with the healthcare Final Year Project. CAD concepts, fabricated work and field-reference images are identified in their respective entries.

ABOUT & EDUCATION

An Industrial Engineering foundation.

I am a Design Engineer with a B.Sc. in Industrial Engineering from UET Peshawar, Jalozai Campus. My practical work covers 3D CAD modeling, mechanical assemblies, engineering drawings, equipment layout and fabrication-oriented projects.

I am interested in translating practical requirements into clear engineering designs, with attention to component placement, manufacturing considerations and system integration. My interests also include lean systems, process improvement and supply chain operations.

B.Sc. Industrial Engineering

UET Peshawar, Jalozai Campus | Graduated 2025 | CGPA 3.29

Mechanical design & CAD

AutoCAD | SolidWorks | PTC Creo | 3D CAD Modeling | Assembly Design | 2D Engineering Drawings | Reverse Engineering | Design for Manufacturing | Equipment Layout & Integration

Manufacturing & documentation

Fabrication Documentation | CNC Machining - Process Exposure | Laser Cutting - Process Exposure | 3D Printing - Process Exposure | MS Word | PowerPoint

Industrial engineering & analysis

Lean & Six Sigma Fundamentals | Value Stream Mapping (VSM) | 5S | Fishbone Analysis | Process Improvement | Excel | Power BI | Inventory Management | Procurement & Supply Chain

PROFESSIONAL JOURNEY

Engineering experience.

December 2025-Present

Tech Valley Pvt Ltd

Design Engineer

Working on engineering design projects involving CAD development, mechanical layouts and component integration. My responsibilities include translating project requirements into models and drawings, developing equipment arrangements and refining designs to suit practical applications. This role continues to strengthen my ability to connect engineering concepts with manufacturing needs while applying process improvement principles throughout design development.

July-November 2025

Robomak Pvt. Ltd.

Design Engineer

Gained hands-on experience in mechanical design using SolidWorks and Creo, developing CAD models and assemblies while building an understanding of how components fit and function together. Learned about manufacturing processes including CNC machining, laser cutting and 3D printing, strengthening my awareness of the relationship between design decisions and fabrication methods.

PROFESSIONAL JOURNEY / CONTINUED

Industrial experience.

August-September 2024

Fauji Fertilizer Company (FFC)

Project Management

Developed an understanding of the project lifecycle, from initiation and planning through execution. Assisted with project planning and milestone development while learning how larger objectives are divided into manageable tasks. Gained exposure to task breakdown, scheduling and the coordination required to organize engineering project activities.

July-August 2024

Pakistan Ordnance Factories

Industrial Layout & Plant Operations

Studied industrial layouts, plant operations and material movement to understand how production activities are organized within a manufacturing environment. Observed the relationship between equipment placement, workflow and operational requirements, connecting Industrial Engineering coursework with practical factory operations.

July-September 2023

Dynea Pakistan Limited - Gadoon Unit

Supply Chain & Store Management

Supported material issuing and order processing while maintaining inventory records and stock control information. Monitored stock levels to help identify potential shortages and excess inventory. This experience developed my understanding of how accurate documentation, material availability and organized store operations support production and supply chain activities.

PROJECT 01 / MECHANICAL

Barrel Cleaning Robot - 125 mm Tank Barrel

A purpose-built barrel cleaning robot developed from an initial concept. Our team worked through the component selection, engineering calculations and mechanical assembly design to create an integrated cleaning system. The project has since progressed into component fabrication and physical assembly work.

Design contribution

Developed the robot from the ground up with the team, defining the mechanical layout and integrating the cleaning and motion-related components into the main assembly.

The work combined design decisions, component selection and engineering calculations rather than adapting a complete existing robot design.

System overview

Two cleaning brushes, positioned at the front and handle-side ends of the assembly.

Two metal-detection sensors used to identify the robot's position relative to the barrel.

Automatic forward and reverse cleaning motion, with reversal when the front brush exits the barrel.

PROJECT 01 / COMPONENT BREAKDOWN

Barrel Cleaning Robot Subassemblies

Wheel-Based Motion Assembly

This subassembly provides movement through the barrel. Three silicone wheels are arranged around the assembly, with their rotating motion moving the robot forward.

The blue housing brings the wheel units together as a compact mechanical assembly within the main robot.

Cam-Driven Brush Mechanism

A cam mechanism produces the brush's repeated cleaning movement. As the mechanism rotates, the brush moves back and then advances with a quick forward stroke.

This repeated back-and-forth action is used to clean the barrel's internal surface.

01.3 / PHYSICAL BUILD

Fabrication & Production

Following the CAD design and subassembly work, our team began production of the Barrel Cleaning Robot. Component fabrication and physical assembly work brought the design into the workshop, as documented in these original photographs.

From design to physical components

The fabrication work produced individual components and progressed into partial mechanical assemblies. The photographs show parts at different stages of the build, including the cleaning brush and assembled mechanical units.

This stage connects the main CAD assembly and the motion and brush subassemblies above with the physical production work.

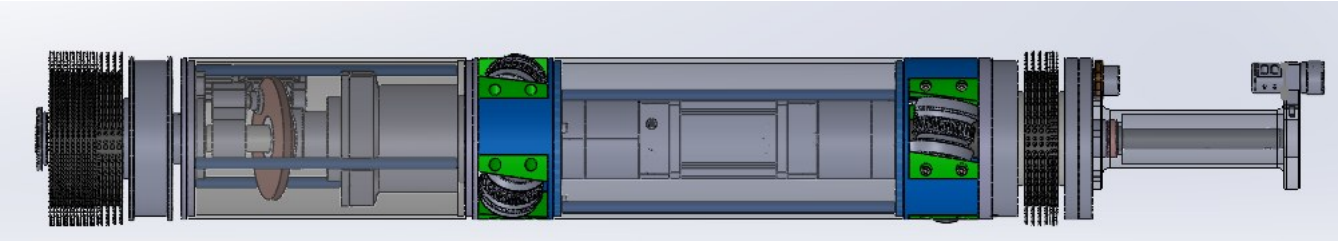
Production & assembly progress

The workshop images document components laid out for assembly and practical assembly work alongside the CAD reference.

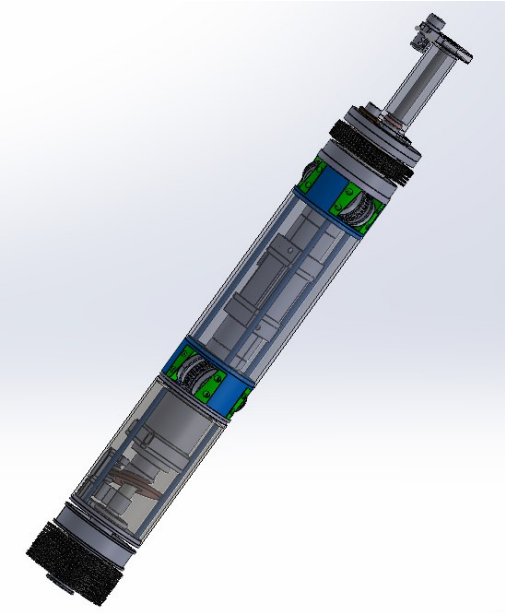
Production has started, with fabrication and assembly progressing at component and subassembly level. These photographs record that work in progress.

PROJECT 01 / VISUAL RECORD

Barrel Cleaning Robot - 125 mm Tank Barrel



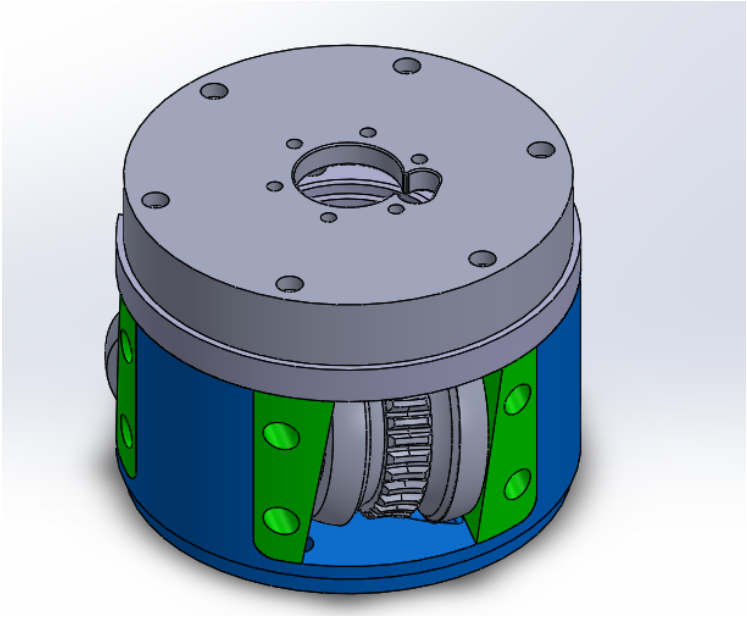
01 / Main assembly · Side view



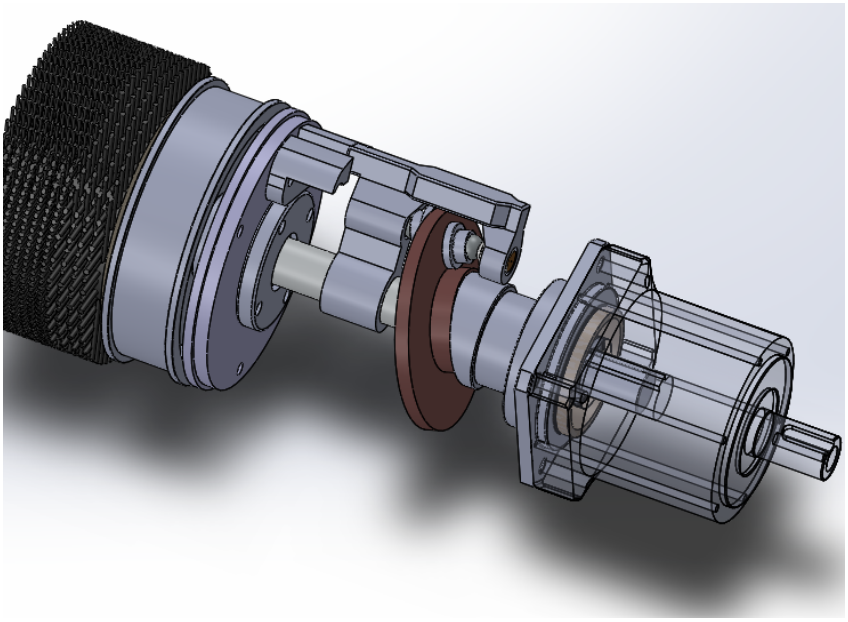
02 / Main assembly · Isometric view

PROJECT 01 / VISUAL RECORD

Barrel Cleaning Robot - 125 mm Tank Barrel



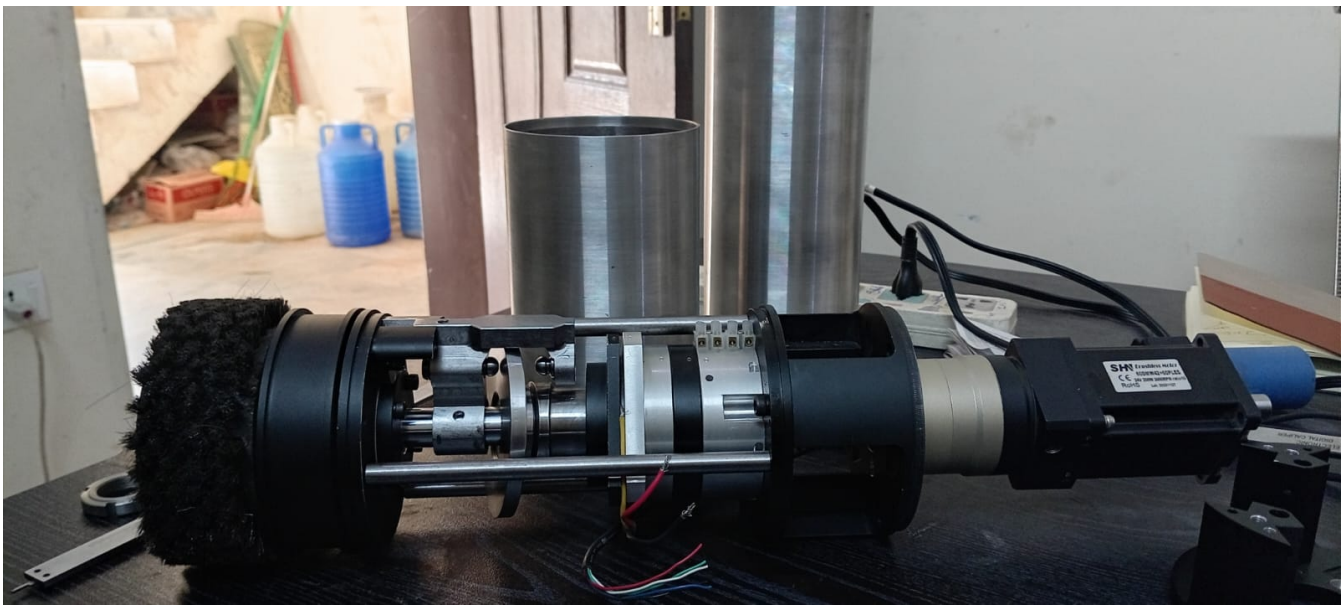
03 / Motion subassembly · CAD assembly view



04 / Brush subassembly · Cam mechanism and cleaning brush

PROJECT 01 / VISUAL RECORD

Barrel Cleaning Robot - 125 mm Tank Barrel



05 / Partial assembly with cleaning brush



06 / Brush and component collection

PROJECT 01 / VISUAL RECORD

Barrel Cleaning Robot - 125 mm Tank Barrel



07 / Workshop subassembly



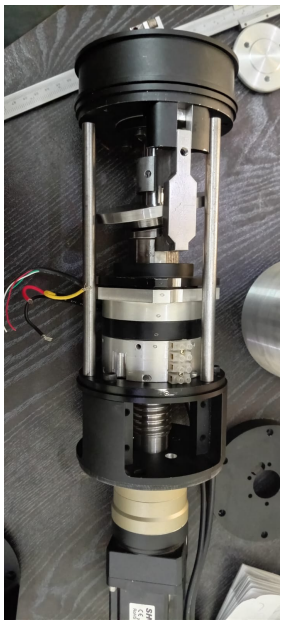
08 / Components laid out for assembly

PROJECT 01 / VISUAL RECORD

Barrel Cleaning Robot - 125 mm Tank Barrel



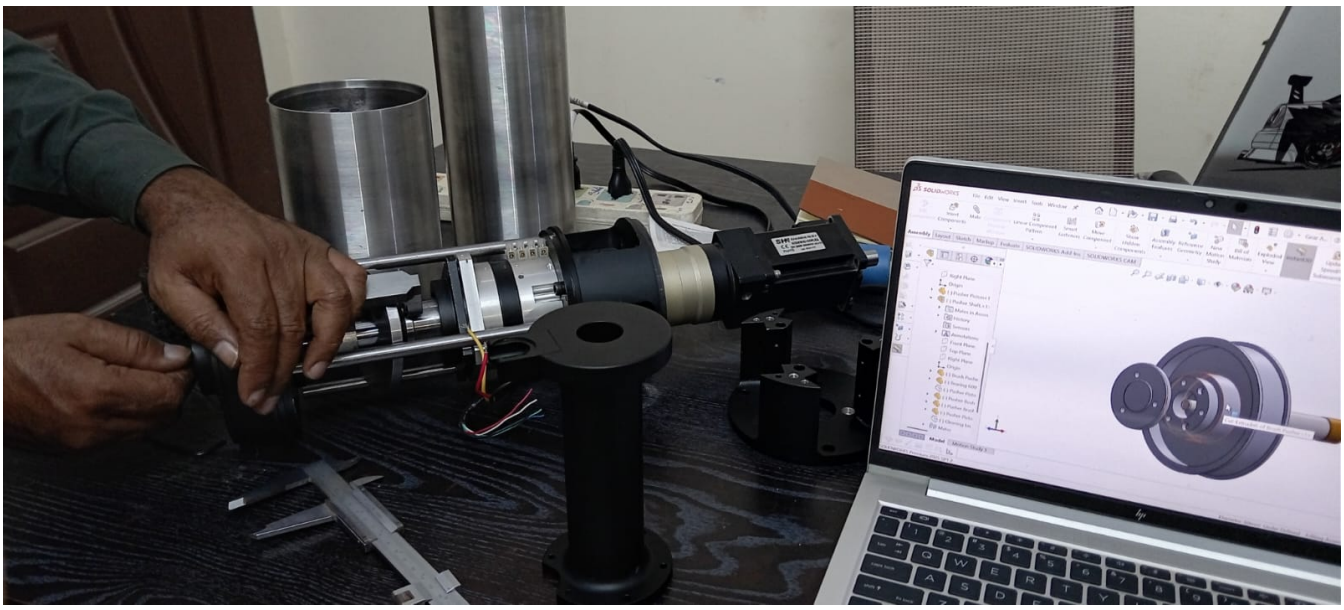
09 / Production-stage component collection



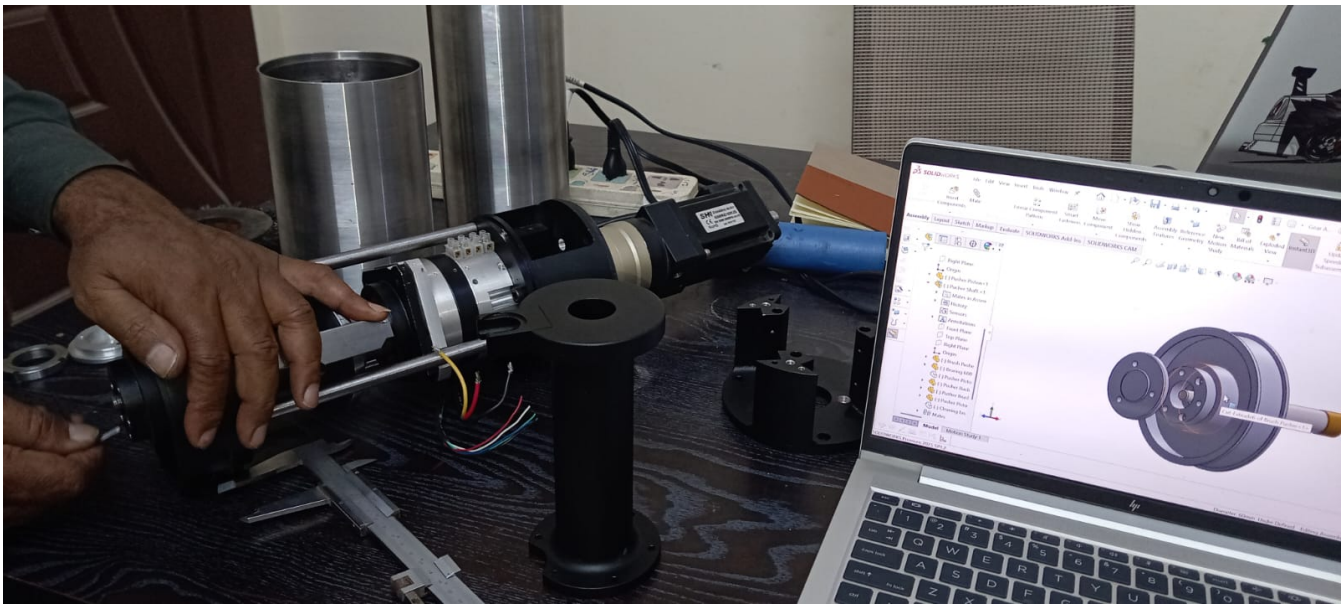
10 / Mechanical subassembly close-up

PROJECT 01 / VISUAL RECORD

Barrel Cleaning Robot - 125 mm Tank Barrel



11 / Workshop assembly alongside CAD



12 / Assembly work and design reference

PROJECT 02 / MECHANICAL

EOD Manipulator

Complete reverse engineering of an Explosive Ordnance Disposal (EOD) manipulator. Our team recreated the mechanical assembly in CAD, representing the overall structure and the gripper used to handle suspicious objects from a distance.

Reverse engineering scope

Our work covered reverse engineering of the complete manipulator, recreating the mechanical design and component arrangement as an integrated assembly.

The project brings together the main structure, handling arm and gripper in a complete CAD representation.

Purpose & application

The manipulator is used in EOD operations to pick up and handle suspicious objects while allowing the operator to remain at a distance.

Its gripper provides the object-handling function during inspection work in operational areas.

PROJECT 02 / COMPONENT BREAKDOWN

EOD Manipulator Subassemblies

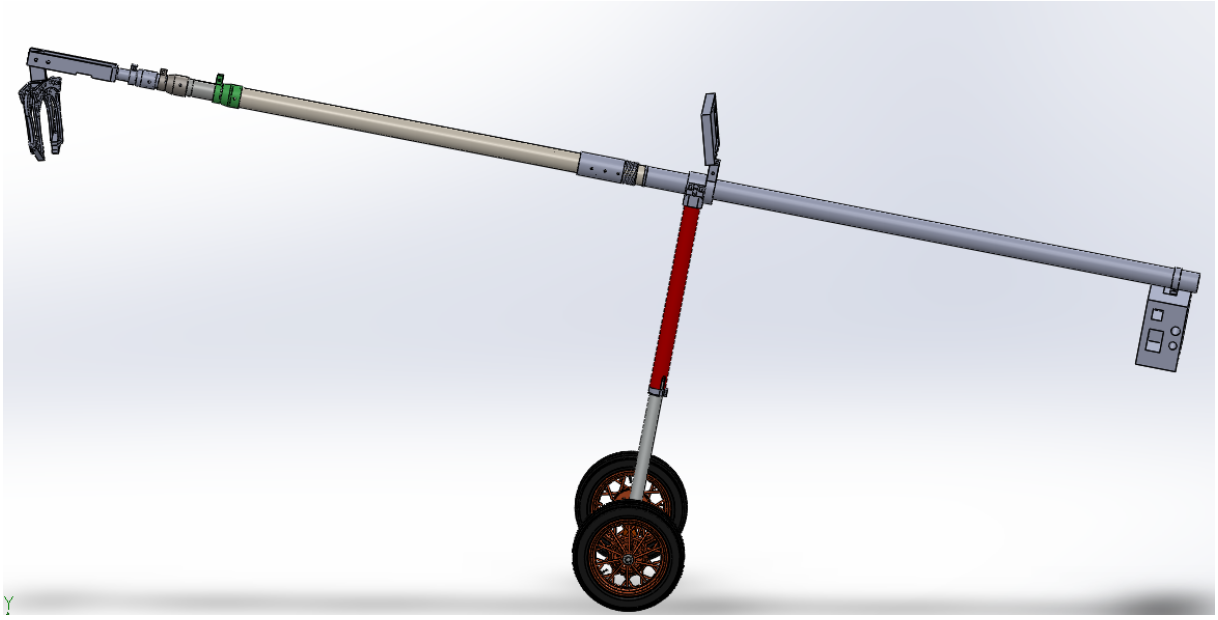
Gripper & Object Handling

The front-end gripper holds objects so they can be picked up and lifted by the EOD manipulator. Sensors are incorporated into the gripper assembly as part of the overall system.

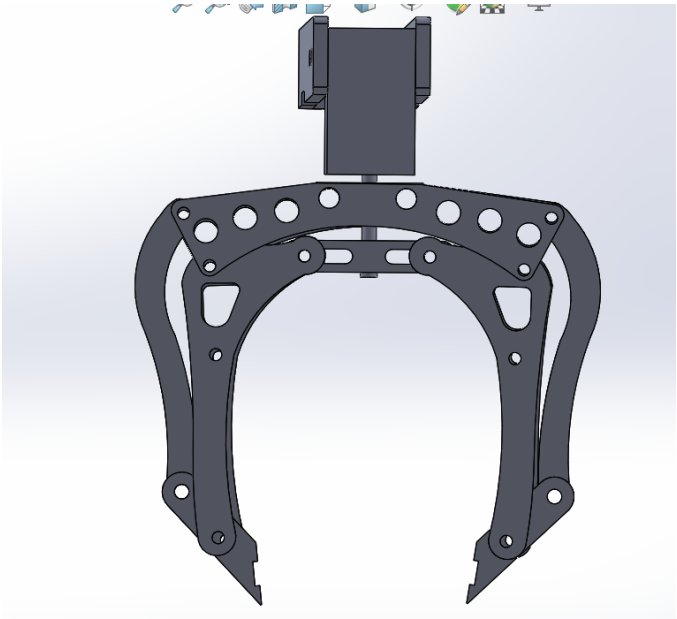
We developed the complete gripper CAD model and assembly as part of the reverse-engineering work. The project also included laser cutting of gripper components, connecting the digital model with physical fabrication.

PROJECT 02 / VISUAL RECORD

EOD Manipulator



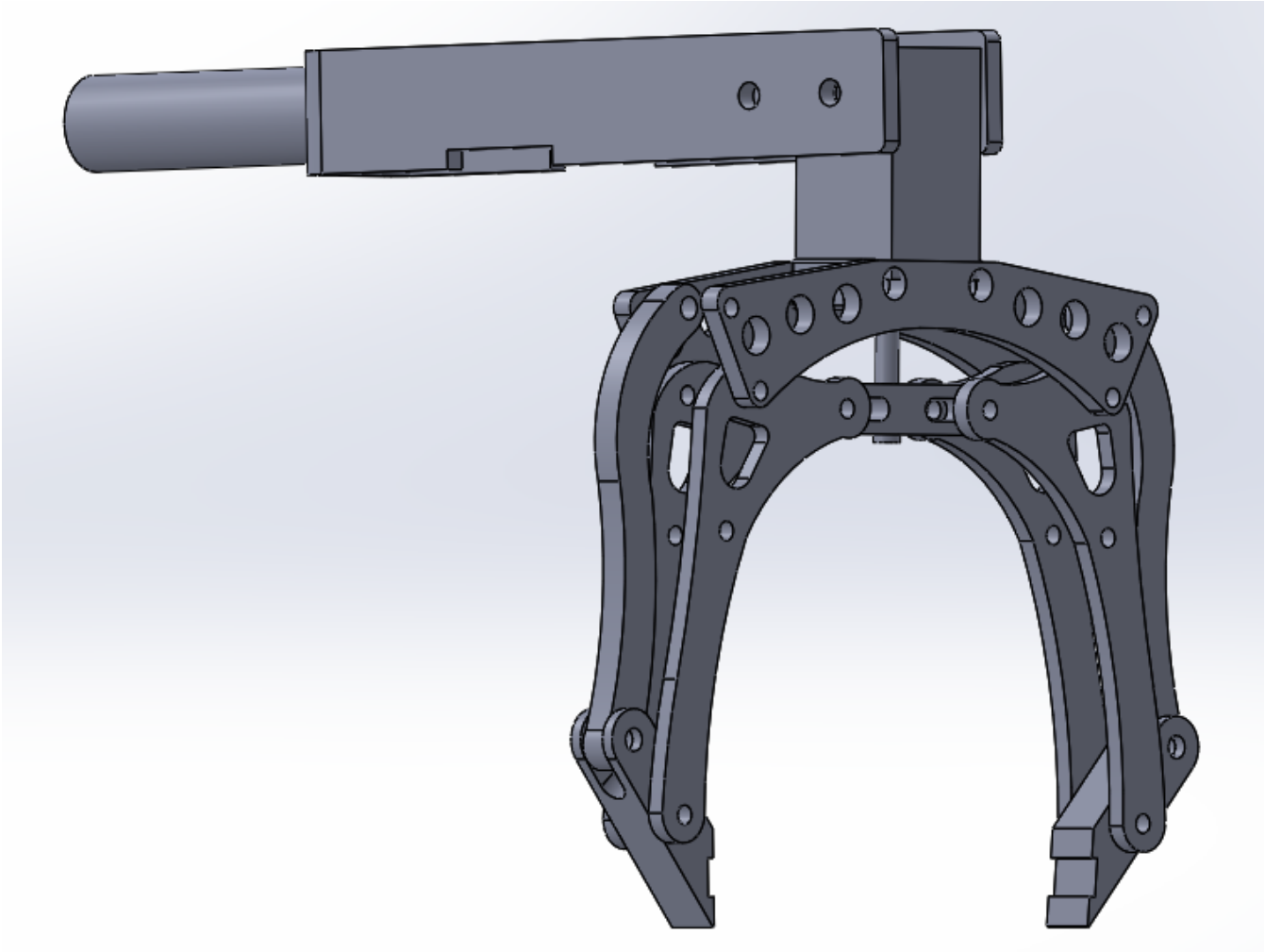
01 / Complete EOD manipulator · Reverse-engineered CAD assembly



02 / Gripper assembly · Front view

PROJECT 02 / VISUAL RECORD

EOD Manipulator



03 / Gripper assembly · Isometric view

PROJECT 03 / MECHANICAL

Rail-Based Drone Launch System

A mechanical launch assembly developed to support a drone's initial transition into flight. The system combines a supporting frame, guide rails and a pushing mechanism within an integrated mechanical assembly.

Project contribution

Our team developed the mechanical assembly, bringing together the frame, rail arrangement and pushing mechanism into a complete CAD model.

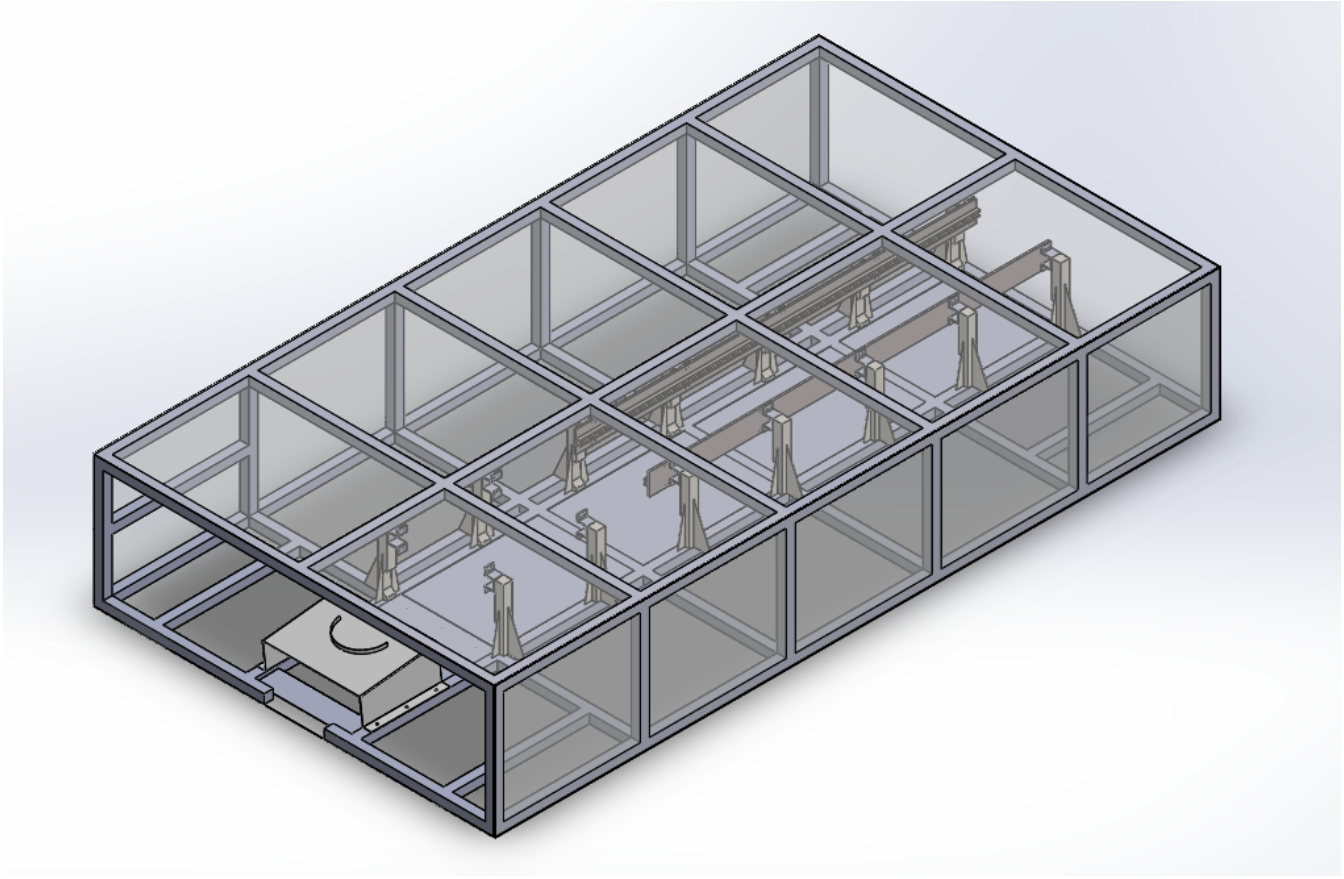
The project focused on the integration of these mechanical components as a drone launch system.

Launch concept

The drone is loaded from the rear of the assembly. A pushing mechanism moves it forward along the rails, providing initial motion before it leaves the launcher and transitions into flight.

PROJECT 03 / VISUAL RECORD

Rail-Based Drone Launch System



01 / Rail-Based Drone Launch System · Overall CAD assembly

PROJECT 04 / COOLING

Shahpar ACU - Mobile AC Truck

Our team built the Shahpar ACU truck for the Shahpar drone project. My work covered PFD preparation, control panel design and fabrication, internal electrical work, integration into the truck body, and the exterior design.

CONTROL PANEL / CAD DEVELOPMENT

My design & build contribution

Prepared the PFD and developed the control panel design, including its enclosure, equipment arrangement and front operating layout. I also fabricated the panel and worked on the truck's internal electrical installation.

My contribution extended to integrating the panel and associated equipment into the truck body, bringing the individual components together within the vehicle assembly.

Body integration & exterior design

I designed the exterior of the AC truck and worked on the equipment integration within its body. The photograph shows the fabricated body, access panels, ventilation openings and exterior graphic treatment.

The generator and associated system controls are brought together at the front control panel, connecting the panel design with the truck's internal installation.

04.1 / FABRICATION & FINISHING

Control Panel Fabrication & Finishing

Panel fabrication

I fabricated the control panel for the AC truck project, taking the panel design through to a physical component. The photograph shows the finished panel with its installed controls and printed front layout.

Surface finishing & printed layout

After fabrication, our team applied powder coating to the panel. Silk-screen printing followed, adding the flow layout, connection lines and equipment labels to the coated surface.

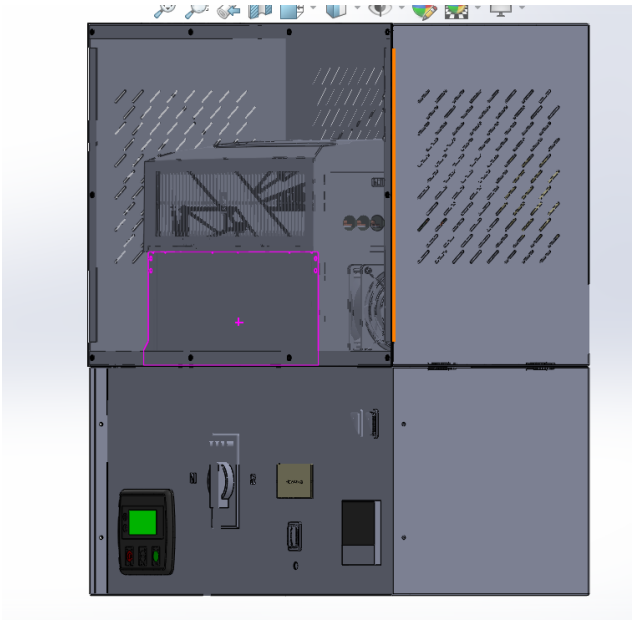
The finished front face brings together the generator controls, changeover switch, monitor, blower control, light switches and HMI in a clearly labelled arrangement.

PROJECT 04 / VISUAL RECORD

Shahpar ACU - Mobile AC Truck



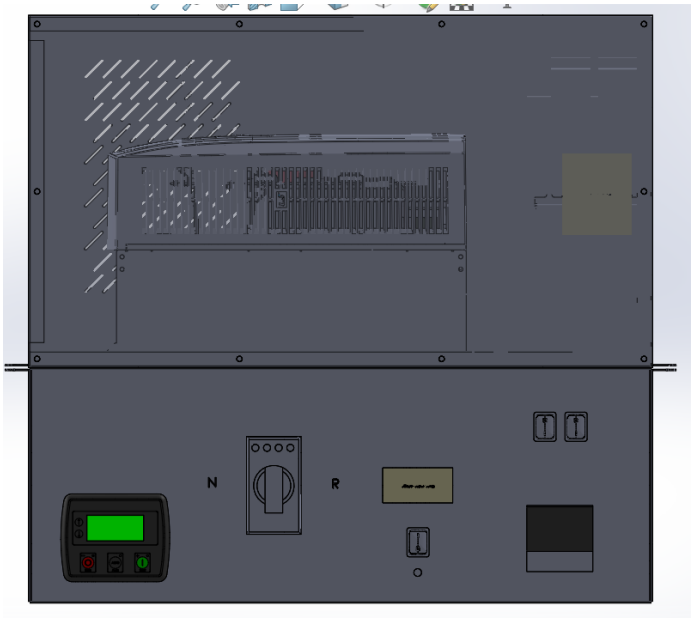
01 / Shahpar ACU · Fabricated truck body and exterior design



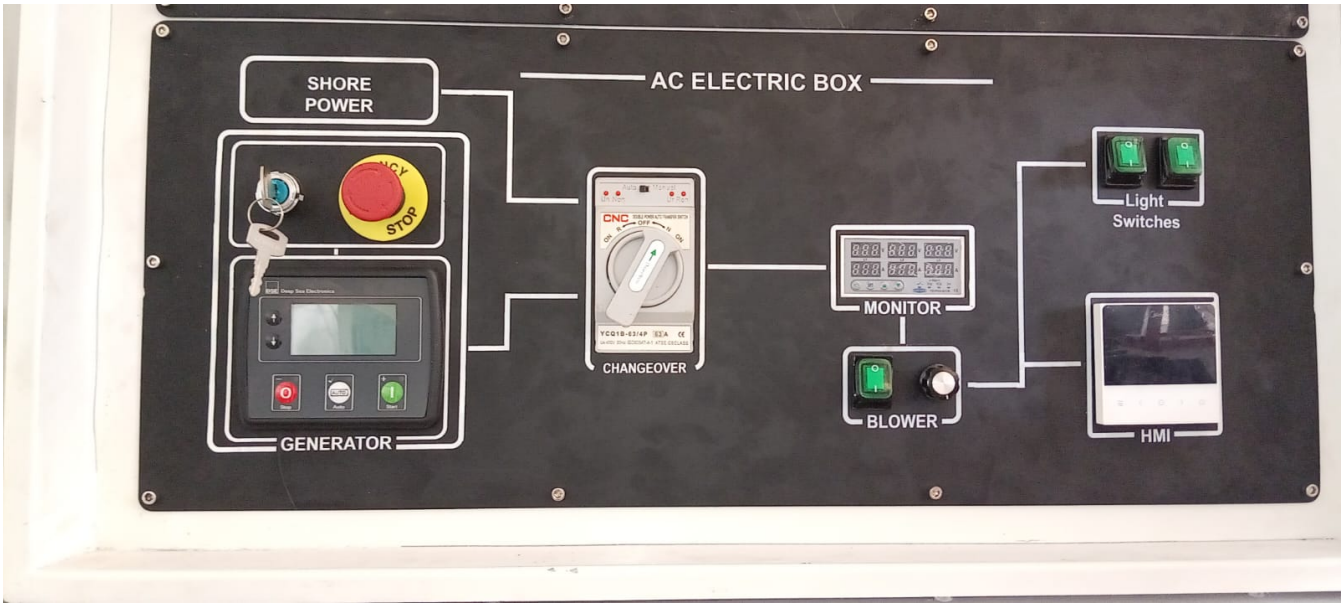
02 / Control panel · Assembly view

PROJECT 04 / VISUAL RECORD

Shahpar ACU - Mobile AC Truck



03 / Control panel · Front view



04 / Fabricated AC control panel · Powder coating and silk-screen printing

PROJECT 05 / VEHICLE

DARK X Radar - Mechanical Design & Integration

FLAGSHIP PROJECT / DEVELOPED FOR NSTP

A manufacturing-focused mechanical design project developed for NSTP, bringing together a radar lifting mechanism and two dedicated cabins. I designed the mechanism for raising and lowering the radar, its operating and control arrangement, and the generator and operator cabins as parts of an integrated CAD assembly.

Radar lifting mechanism & controls

My contribution covered the complete mechanical lifting arrangement used to move the radar between its lower and raised positions. I also developed the mechanism's operating and control arrangement as part of the overall design.

The two CAD views show the assembly in these different positions, presenting the lifting system within the wider mechanical layout.

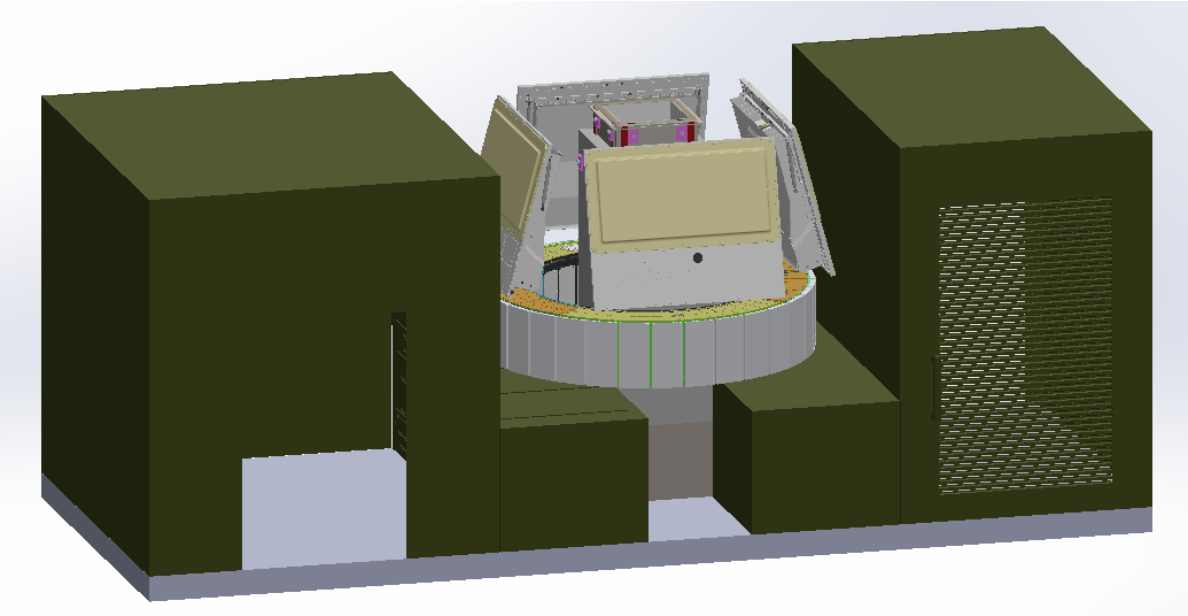
Generator & operator cabins

I designed two separate cabins alongside the radar assembly: one for the generator and another for the operator responsible for operating the radar.

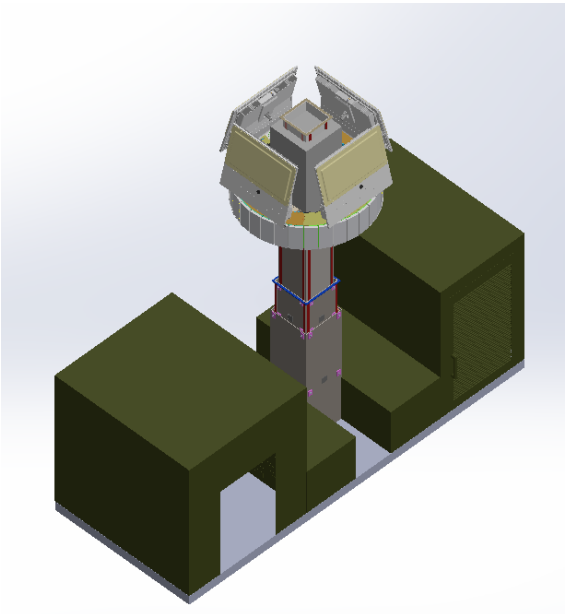
The cabin design and lifting assembly were brought together in a manufacturing-focused layout, combining mechanism design, supporting structures and equipment accommodation in one project.

PROJECT 05 / VISUAL RECORD

DARK X Radar - Mechanical Design & Integration



01 / DARK X · Lowered radar assembly view



02 / DARK X · Raised radar assembly view

PROJECT 06 / VEHICLE

Type 84 Transport Vehicle Launcher - Cabin & Vehicle Design

I developed the cabin manufacturing design, rear access stairs and rear lighting arrangement for a Type 84 transport vehicle launcher project. My work included CAD assembly design and 2D engineering drawings documenting these contributions within the wider vehicle layout.

My design contribution

My responsibility covered the cabin's manufacturing design, the rear stairs and the rear lighting arrangement. I prepared the CAD representation and supporting 2D drawings for this scope.

The project highlights my work on vehicle cabin design, access components and drawing documentation with manufacturing and assembly requirements in mind.

Project context

The team worked with a vehicle sourced from China and incorporated an existing Type 84 launcher at the rear. My cabin design formed part of this broader vehicle integration project.

The isometric and side views present the cabin in the context of the overall CAD assembly.

06.1 / DESIGN DOCUMENTATION

2D Engineering Drawings

Drawing documentation

These selected 2D drawings complement the CAD assembly views above. They show the overall vehicle arrangement and the rear platform, stairs and lighting layout, including multiple drawing views and a parts list.

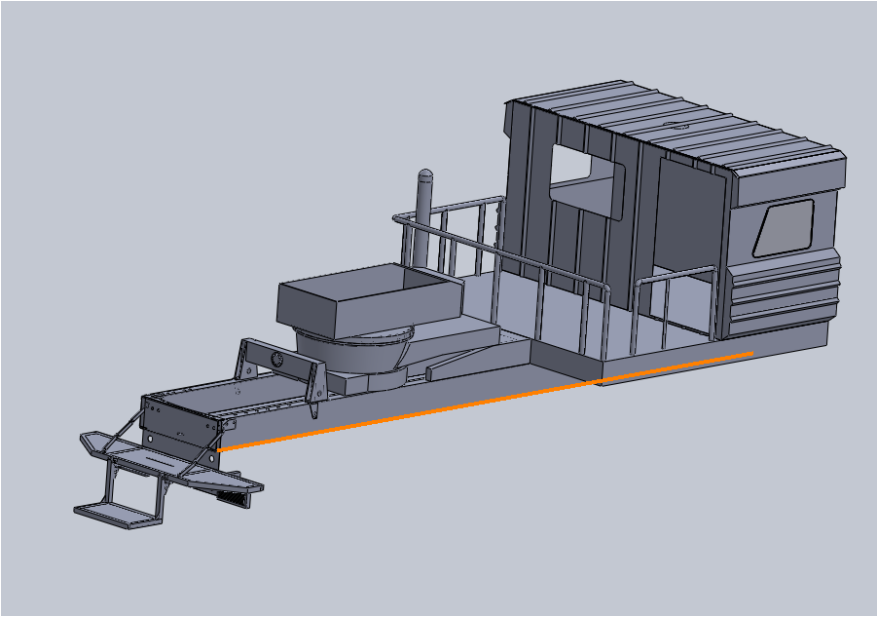
My design scope included the cabin, rear access stairs and rear lighting arrangement.

Confidentiality & selected work

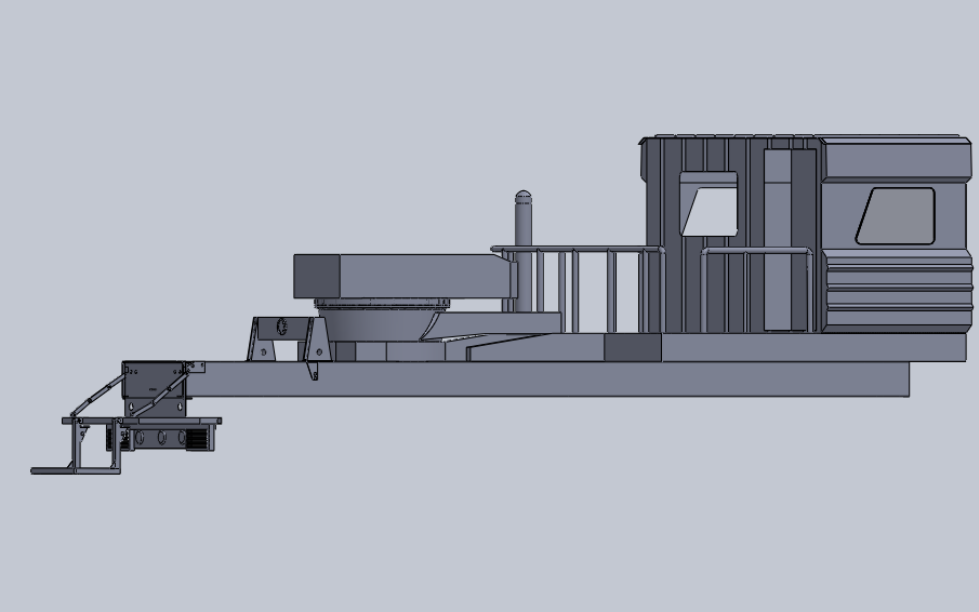
Selected CAD views and 2D drawings are presented to illustrate my design contribution. Complete 3D models and restricted project details are withheld due to confidentiality.

PROJECT 06 / VISUAL RECORD

Type 84 Transport Vehicle Launcher - Cabin & Vehicle Design



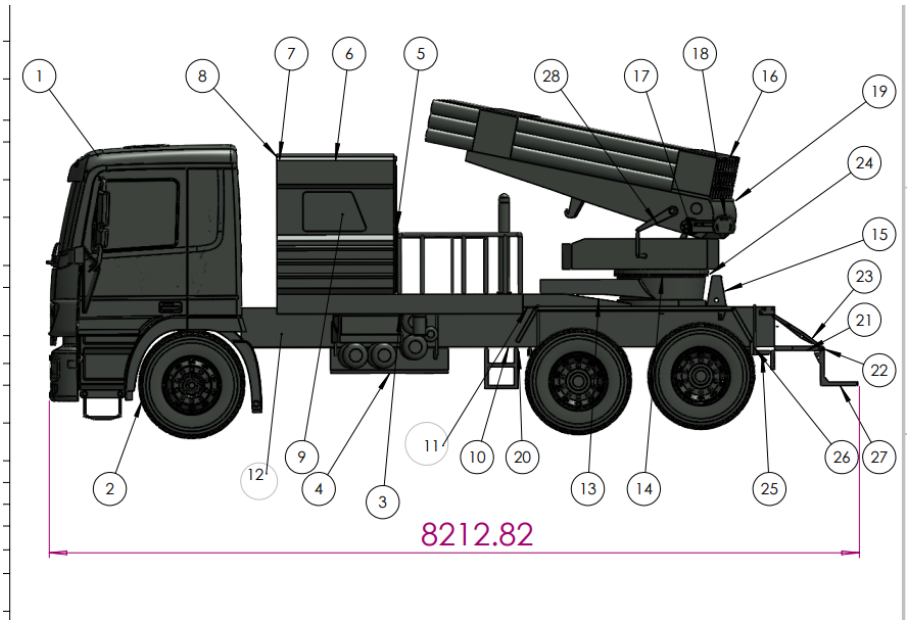
01 / Cabin and platform assembly · Isometric view



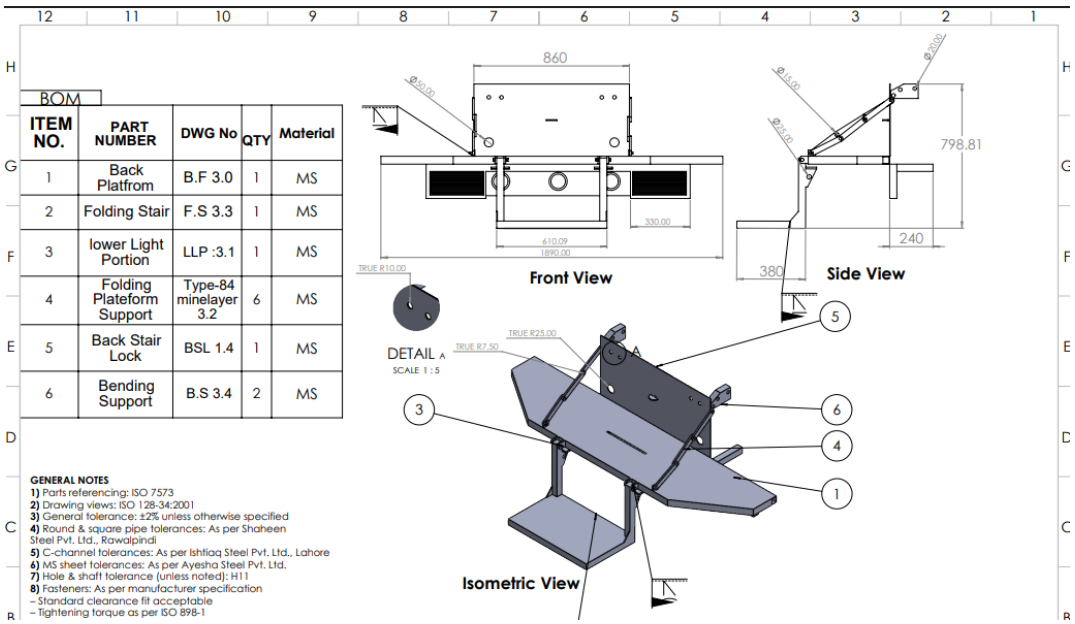
02 / Cabin and platform assembly · Side view

PROJECT 06 / VISUAL RECORD

Type 84 Transport Vehicle Launcher - Cabin & Vehicle Design



03 / Selected 2D drawing · Vehicle assembly overview



04 / Selected 2D drawing · Rear stairs and lighting arrangement

PROJECT 07 / MECHANICAL

Custom 41U Vehicle Equipment Rack

RECENTLY COMPLETED DESIGN

A custom 41U equipment rack developed for a vehicle application. I completed the mechanical CAD design and assembly, working to standard rack requirements while adapting the enclosure and overall arrangement to the project's specific needs.

Custom design contribution

Developed the rack's mechanical layout as a complete CAD assembly, bringing together the enclosure frame, vertical equipment-mounting rails and panel arrangement.

The work focused on a 41U rack tailored to the intended vehicle application, balancing the required rack format with a project-specific design.

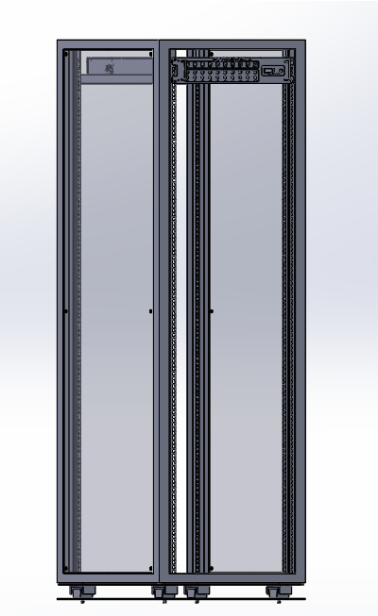
Standard format, tailored assembly

Standard rack requirements guided the design, while the overall enclosure and component arrangement were customized for the project.

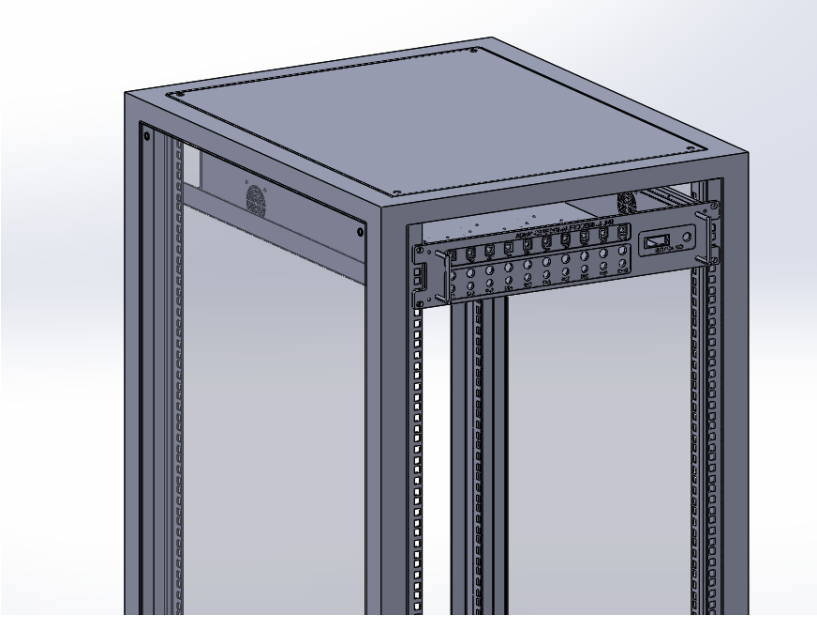
The full-height view presents the complete rack, and the upper detail shows the relationship between the enclosure, mounting rails and equipment panel.

PROJECT 07 / VISUAL RECORD

Custom 41U Vehicle Equipment Rack



01 / Custom 41U rack · Full assembly view



02 / Custom 41U rack · Upper assembly detail

PROJECT 08 / VEHICLE

Pickup-Mounted Operator Cabin Concept

CONCEPT DESIGN / SELECTED 2D DOCUMENTATION

A concept design exploring how a pickup's cargo bed could accommodate an enclosed operator workspace. I developed a cabin layout bringing together seating, a computer workstation and air conditioning within the vehicle-mounted enclosure.

Cabin & workstation concept

Developed the cabin concept around the pickup's rear cargo area, considering how the enclosure, seating and computer workstation could be arranged together.

An air-conditioning unit was incorporated into the proposed layout. The drawing presents the relationship between the vehicle, cabin and interior workspace.

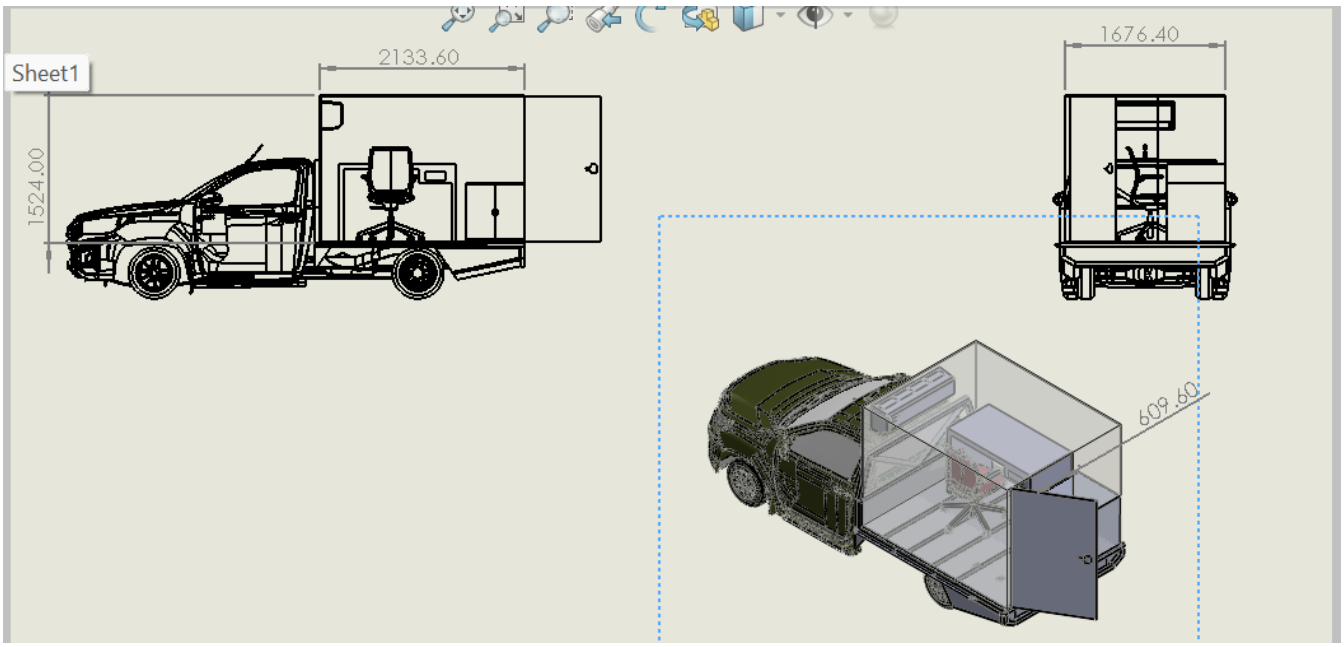
Concept presentation & confidentiality

This project represents a design concept for a mobile workspace, presented through a selected 2D drawing sheet.

The complete 3D model is not shared in this portfolio due to project confidentiality. The selected drawing communicates the concept and my layout-design contribution.

PROJECT 08 / VISUAL RECORD

Pickup-Mounted Operator Cabin Concept



01 / Pickup-mounted operator cabin · Selected 2D concept drawing sheet

PROJECT 09 / VEHICLE

Crane-Assisted Drone Box Transport Vehicle

REVERSE ENGINEERING / CAD CONCEPT

A reverse-engineered CAD vehicle concept combining a truck-mounted crane with a transport platform for drone storage boxes. Our team developed the digital assembly to illustrate how the boxes could be lifted and transferred between locations as part of a mobile handling concept.

Reverse-engineering contribution

Developed the vehicle model through reverse-engineering work, representing the truck, crane and storage-box arrangement within one CAD assembly.

The concept presents the crane and carrying platform together, showing the proposed vehicle layout for moving drone storage boxes.

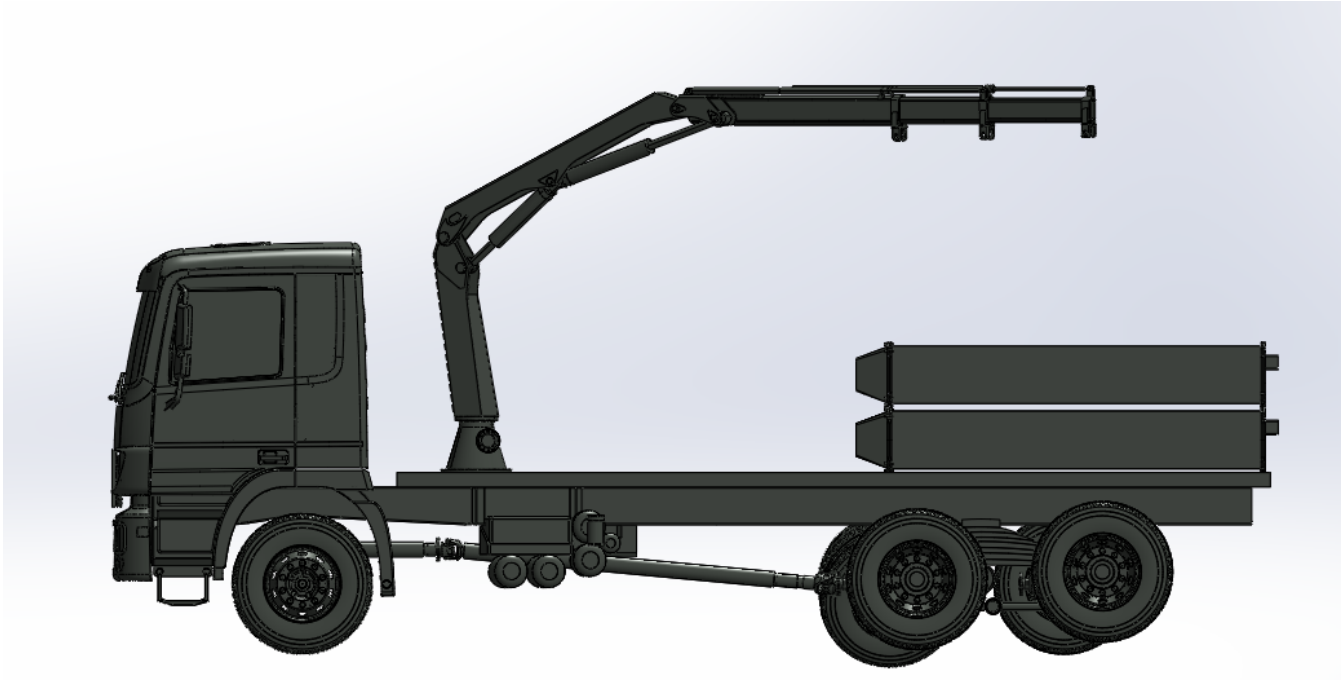
Handling concept & project stage

The crane is intended to lift the storage boxes so they can be transferred from one position to another. The project explores this handling function at the vehicle-concept level.

This project remains a CAD model only. It has not been manufactured.

PROJECT 09 / VISUAL RECORD

Crane-Assisted Drone Box Transport Vehicle



01 / Crane-assisted transport vehicle · Reverse-engineered CAD concept

PROJECT 10 / VEHICLE

NSTP Alpha 12 - Multi-Vehicle 3D Modeling

NSTP ALPHA 12 / THREE VEHICLE PLATFORMS

Developed detailed 3D CAD models for three different vehicle platforms under the NSTP Alpha 12 project. I personally created the assemblies and delivered the model files to NSTP for subsequent development, focusing on design accuracy, component integration and practical vehicle layouts.

Three-vehicle CAD modeling

Created the 3D designs and detailed vehicle assemblies using CAD tools, representing the overall vehicle forms and the arrangement of cabins, enclosures and associated components.

The work emphasized consistency between the reference vehicles and the digital assemblies, with attention to component integration and real-world feasibility as design objectives.

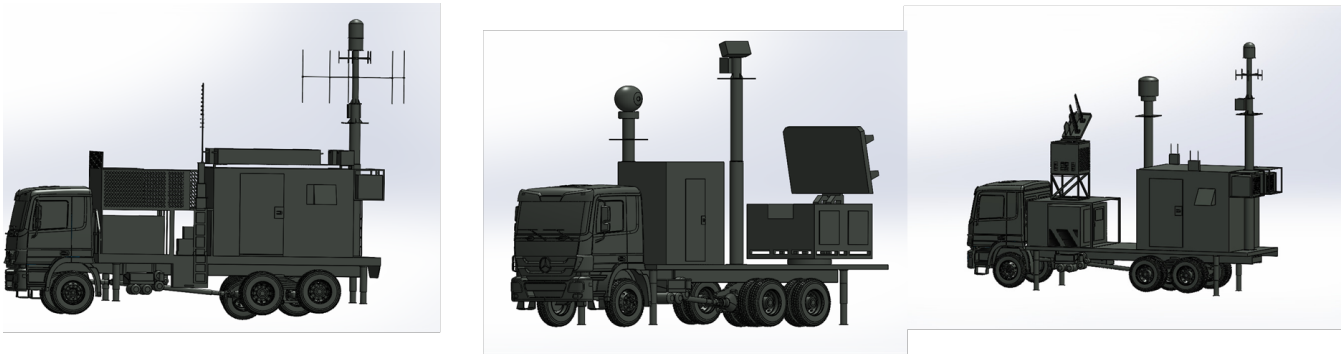
Delivered design files & project scope

The assignment was to provide the 3D models for NSTP's later vehicle development work. My deliverable was the completed CAD model files.

The field photograph is included as reference context. The CAD views show my modeling work, which strengthened my experience in complex assemblies and mechanical system design.

PROJECT 10 / VISUAL RECORD

NSTP Alpha 12 - Multi-Vehicle 3D Modeling



01 / Three vehicle platforms · CAD assembly overview



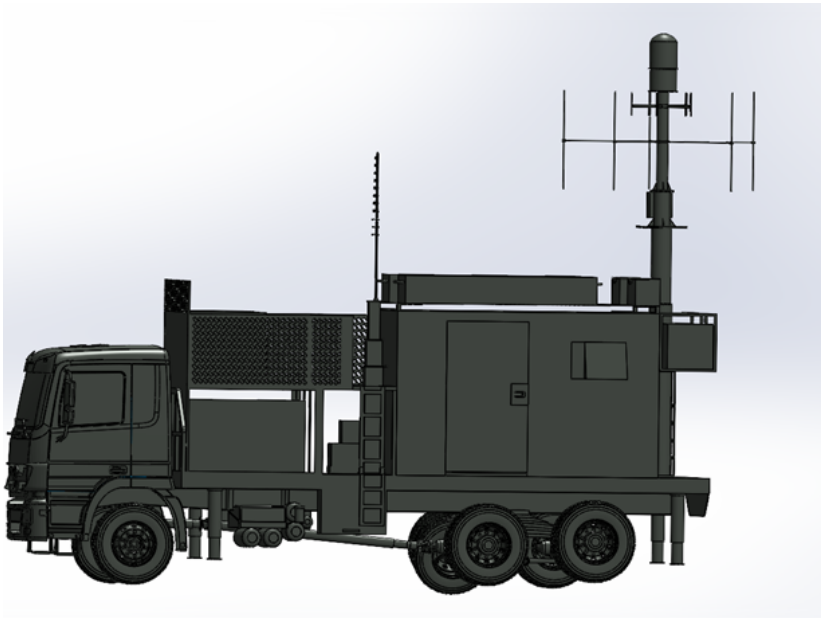
02 / Field reference photograph · Modeling context

PROJECT 10 / VISUAL RECORD

NSTP Alpha 12 - Multi-Vehicle 3D Modeling



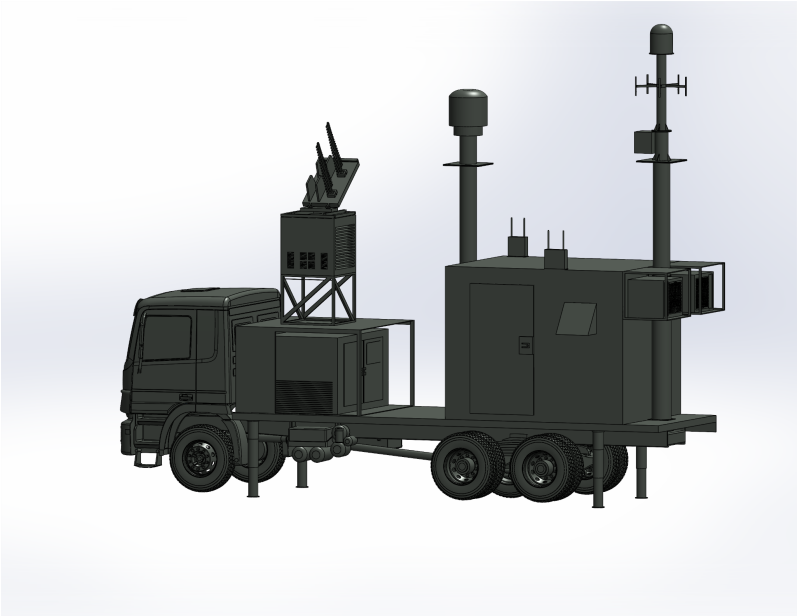
03 / Vehicle CAD assembly · Isometric overview



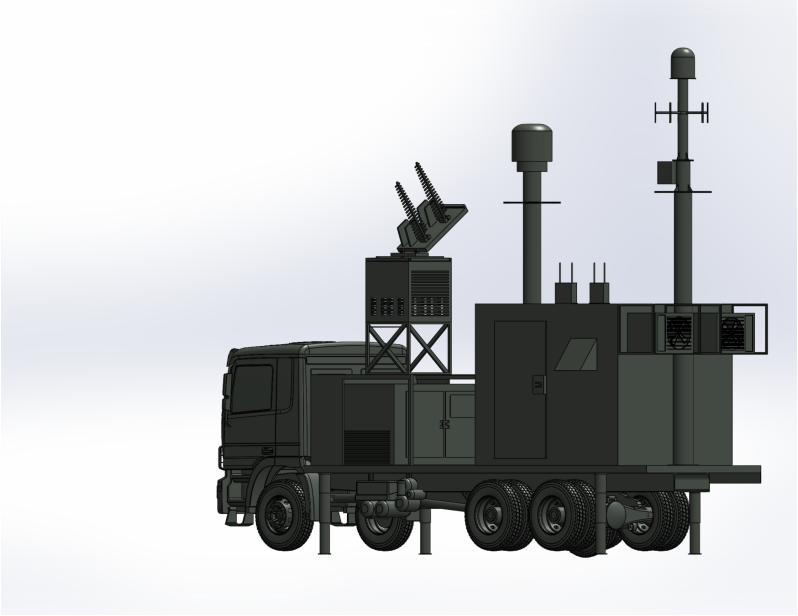
04 / Vehicle CAD assembly · Side overview

PROJECT 10 / VISUAL RECORD

NSTP Alpha 12 - Multi-Vehicle 3D Modeling



05 / V-1 model · Detailed CAD view



06 / V-1 model · Alternate CAD view

PROJECT 11 / MECHANICAL

Cabin-Integrated Vertical Lifting System

3D CAD CONCEPT / DESIGN PRESENTATION

A 3D CAD concept combining a central vertical lifting arrangement with adjacent cabins. The model demonstrates a proposed system in which equipment can be raised for use and lowered again once the task is complete.

Concept modeling contribution

Developed the 3D CAD assembly to bring together the cabins, supporting frame and central lifting arrangement within one proposed layout.

The model was created to communicate the design idea and help others visualize how the overall system could be arranged.

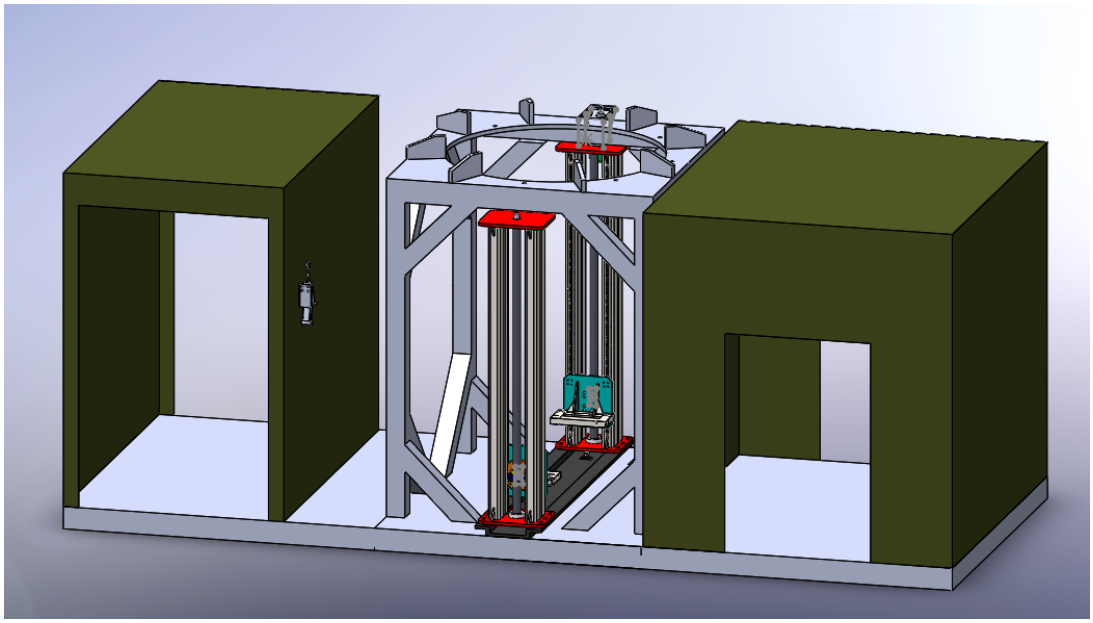
Proposed movement & project stage

The concept allows equipment placed within the central lifting arrangement to move upward for use, then return to a lower position when the work is finished.

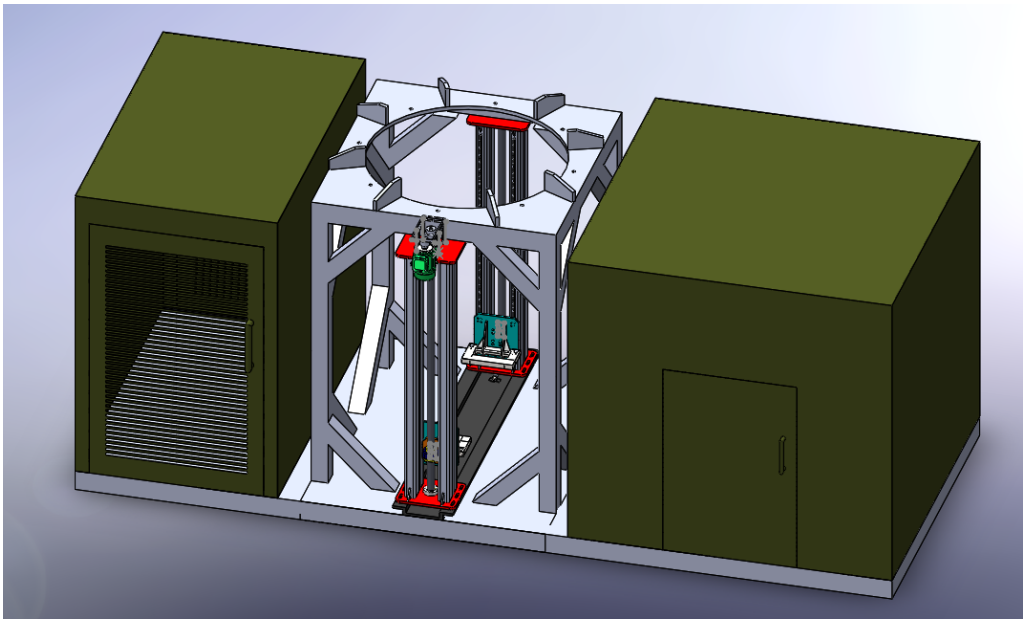
This project is a CAD concept for presentation purposes. It has not been manufactured.

PROJECT 11 / VISUAL RECORD

Cabin-Integrated Vertical Lifting System



01 / Vertical lifting system · Overall CAD concept



02 / Vertical lifting system · Alternate assembly view

PROJECT 12 / VEHICLE

Mobile Workshop Vehicle

VEHICLE DESIGN & FABRICATION / AZERBAIJAN

A vehicle-based workshop project combining mechanical design with practical fabrication. I developed the vehicle design and internal equipment layout, including the placement of a lathe, cylinders and other workshop equipment. Our team fabricated the vehicle in Azerbaijan.

Vehicle & equipment layout

Designed the workshop body and planned the placement of the lathe, cylinders and associated equipment within the available vehicle space.

The CAD assembly brings the workshop enclosure and equipment arrangement together, showing how the individual components fit within the mobile workspace.

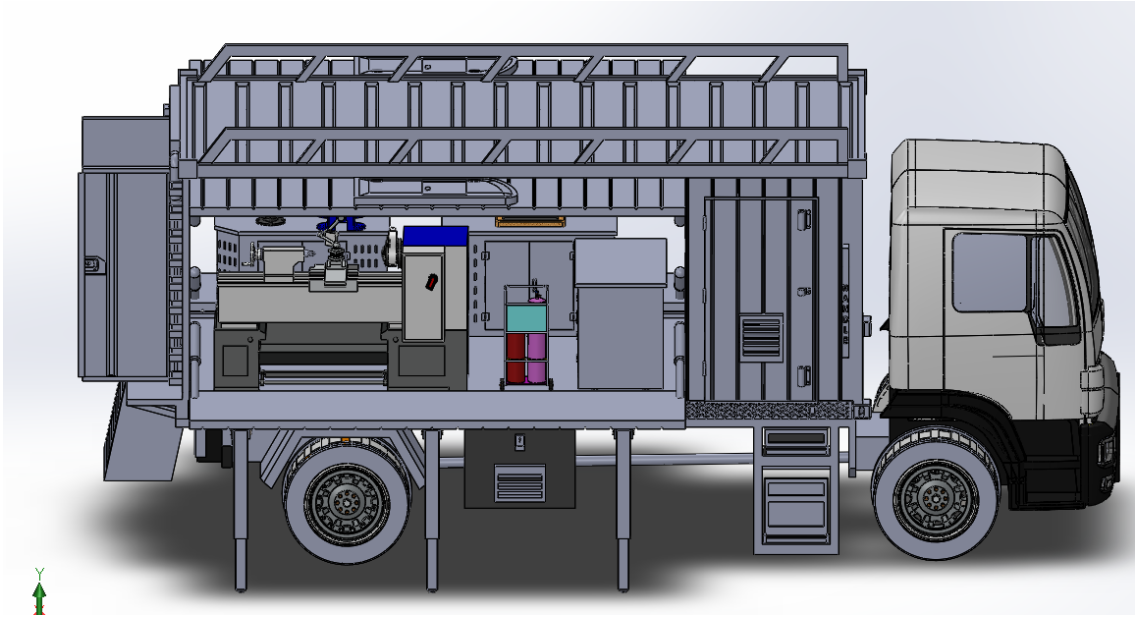
Fabrication in Azerbaijan

Our team fabricated the mobile workshop vehicle in Azerbaijan, taking the project from the design stage into a physical build.

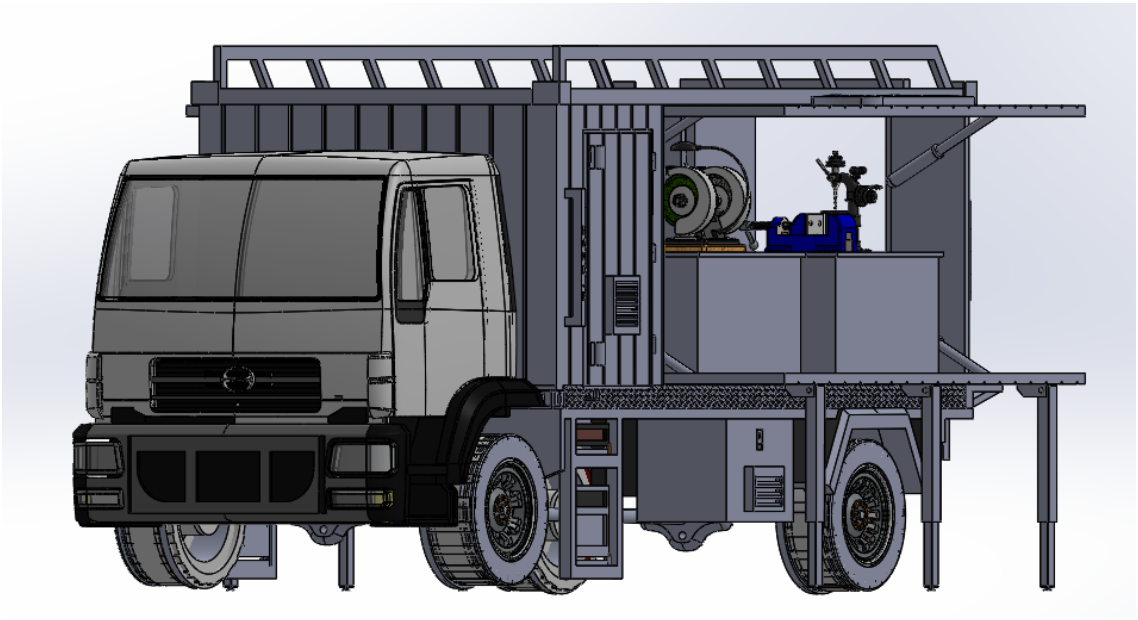
This project connects my CAD and equipment-layout work with practical vehicle fabrication. The images shown here are CAD views of the workshop design.

PROJECT 12 / VISUAL RECORD

Mobile Workshop Vehicle



01 / Mobile workshop · Internal equipment layout in CAD



02 / Mobile workshop · Overall CAD assembly view

PROJECT 13 / VEHICLE

Aviation Bowser

3D CAD & ENGINEERING DOCUMENTATION

Developed the aviation bowser's 3D CAD model and prepared the complete set of 2D engineering drawings, completing outstanding design documentation. I also coordinated the drawings' review through RDE and obtained clearance before the documentation was forwarded for the next stage.

3D model & complete drawing set

Created the vehicle's 3D model and completed the 2D engineering drawing package. The work addressed the project's incomplete documentation and brought the design into a coordinated set of CAD deliverables.

The assembly view presents the overall vehicle layout, including the tank body, equipment enclosure and access platform.

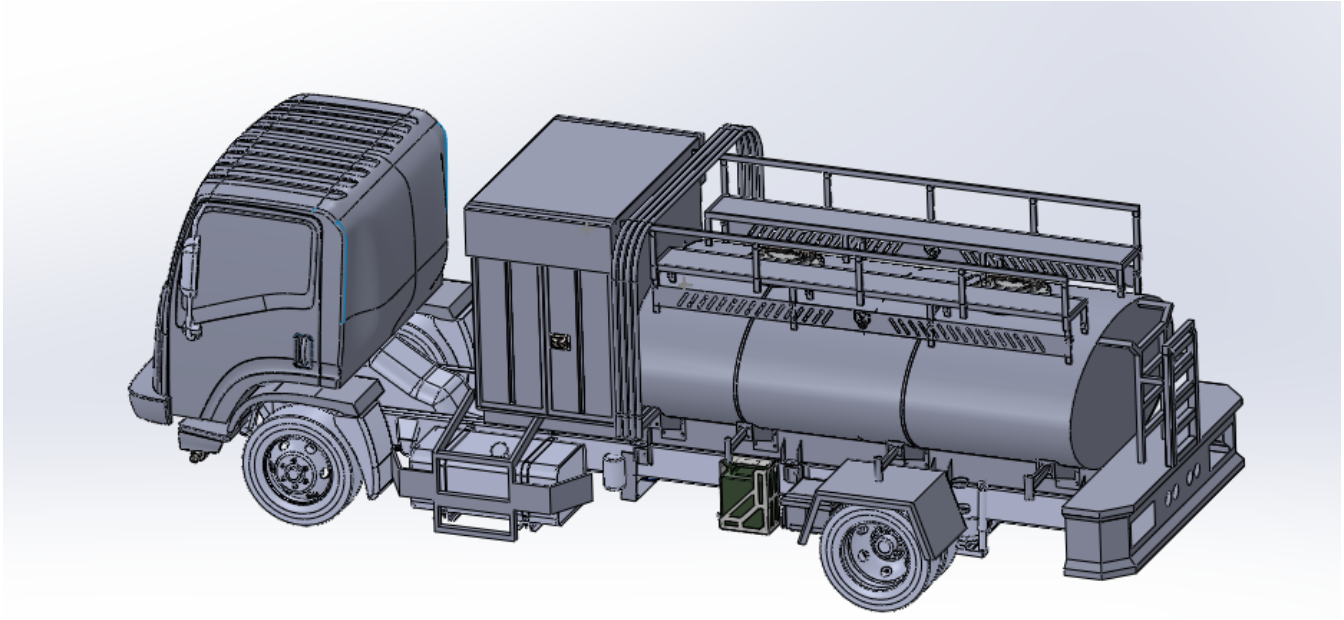
RDE review & onward submission

Managed the drawing submission through RDE's review process and obtained clearance for the engineering documentation.

Following clearance, the documentation was forwarded for the next stage, completing my contribution from CAD modeling and drawing preparation through review coordination.

PROJECT 13 / VISUAL RECORD

Aviation Bowser



01 / Aviation bowser · Overall 3D CAD assembly

PROJECT 14 / COOLING

AC Truck - 16-Ton Cooling System

Developing a truck-mounted, 16-ton air-conditioning system for large-engine cooling. The design brings the cooling equipment, high-capacity blower, ducting and supporting components together in a coordinated mobile arrangement.

My design contribution

My work covers the overall mechanical design and equipment layout, including placement of the outdoor unit, air ducts and variable-frequency drive (VFD). I am also organizing the generator within the vehicle arrangement so the major components can be presented as one integrated system.

The CAD assembly communicates the relationship between the cooling unit, blower and duct routing, supporting further design development and placement decisions.

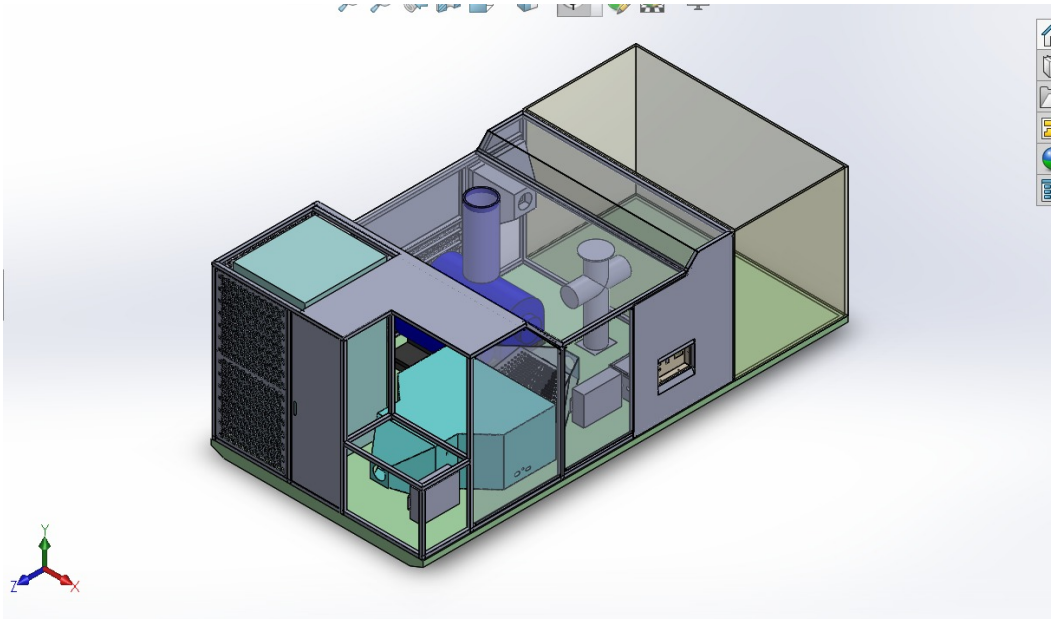
System purpose & project stage

The system is intended to deliver cooling air to the required engine location through its ducting. The focus of this portfolio entry is the equipment arrangement and mechanical design work currently in progress.

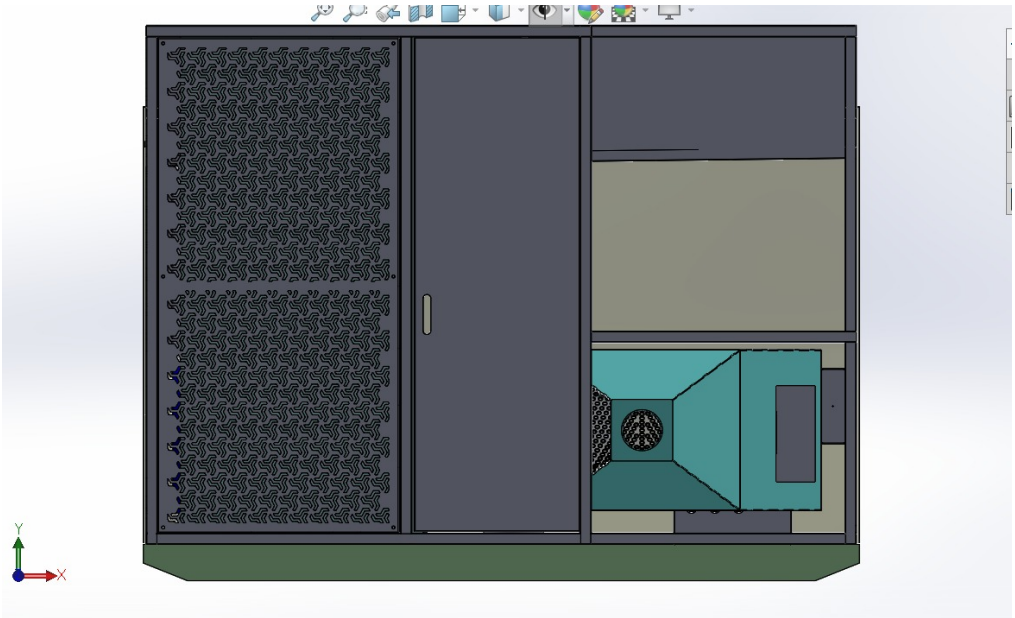
The gallery contains four original CAD views of the proposed arrangement.

PROJECT 14 / VISUAL RECORD

AC Truck - 16-Ton Cooling System



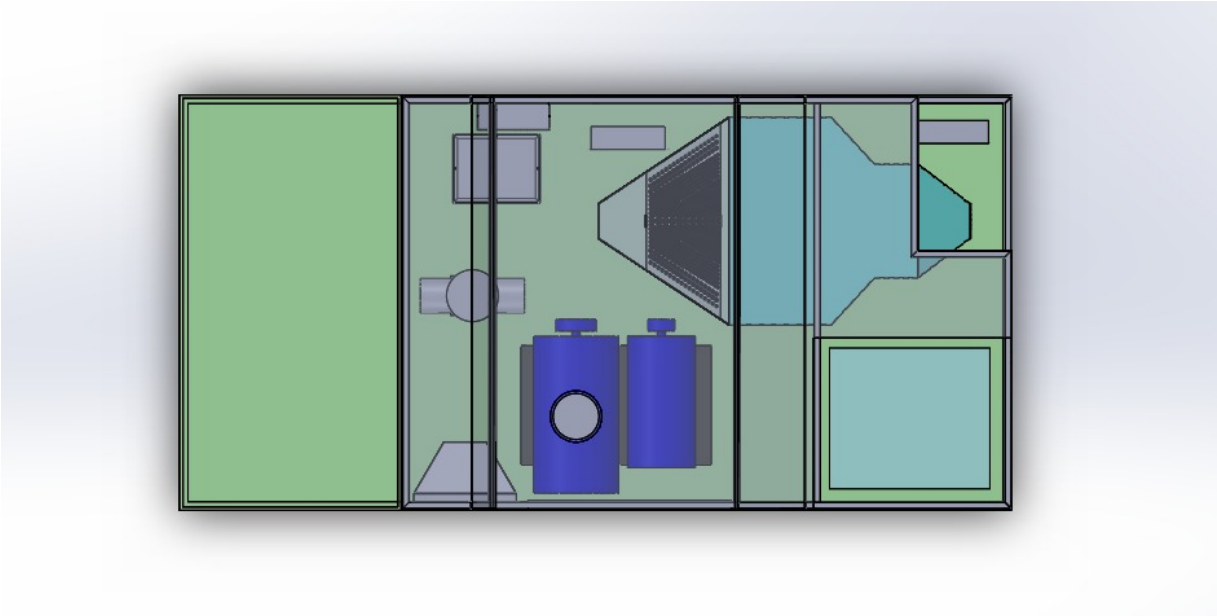
01 / Overall CAD arrangement



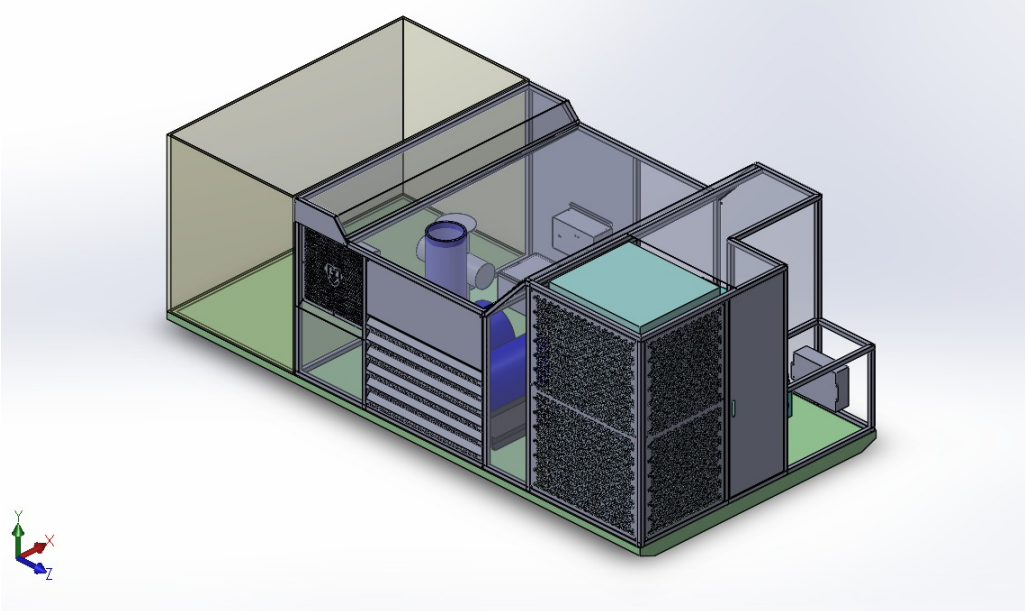
02 / Front equipment layout

PROJECT 14 / VISUAL RECORD

AC Truck - 16-Ton Cooling System



03 / Top-view component placement



04 / Alternate CAD assembly view

PROJECT 15 / COOLING

Dragon AC Cooling Cart

A compact mobile cooling cart developed for aircraft and engine cooling. The project combines a 3D CAD assembly with a fabricated physical model, showing the progression from equipment arrangement to a workshop build.

CAD design & equipment arrangement

Developed the cooling-cart model as an integrated mechanical assembly, bringing the equipment enclosure and trolley arrangement into one design. The selected isometric, side and front views show the overall layout and how the components occupy the available space.

This smaller unit is presented separately from the 16-ton AC Truck, reflecting its own mechanical arrangement and mobile cooling application.

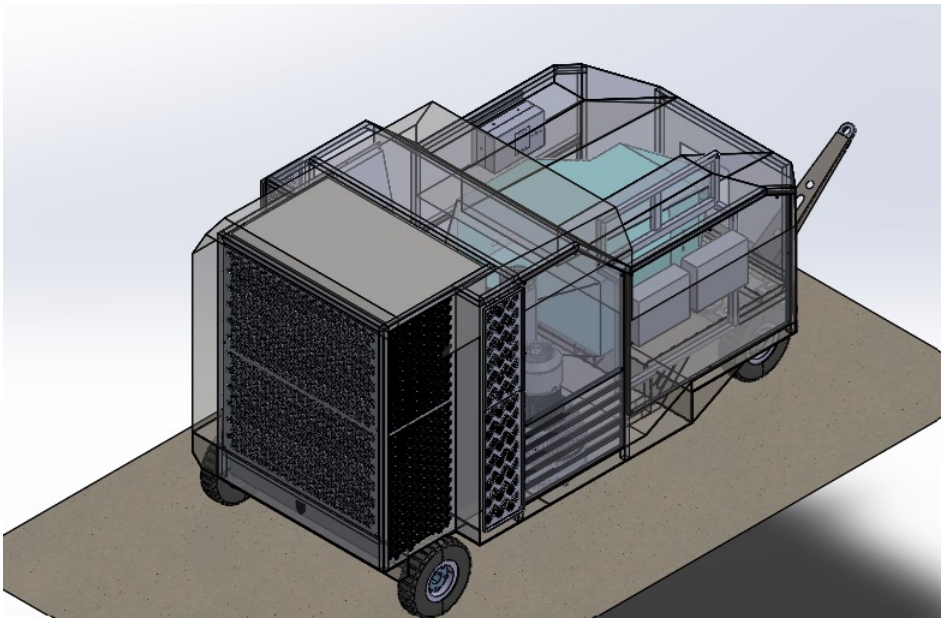
Fabrication evidence

A physical model was fabricated, connecting the CAD work with a real workshop outcome. The final gallery image shows the fabricated cart, while the first three images document the 3D design.

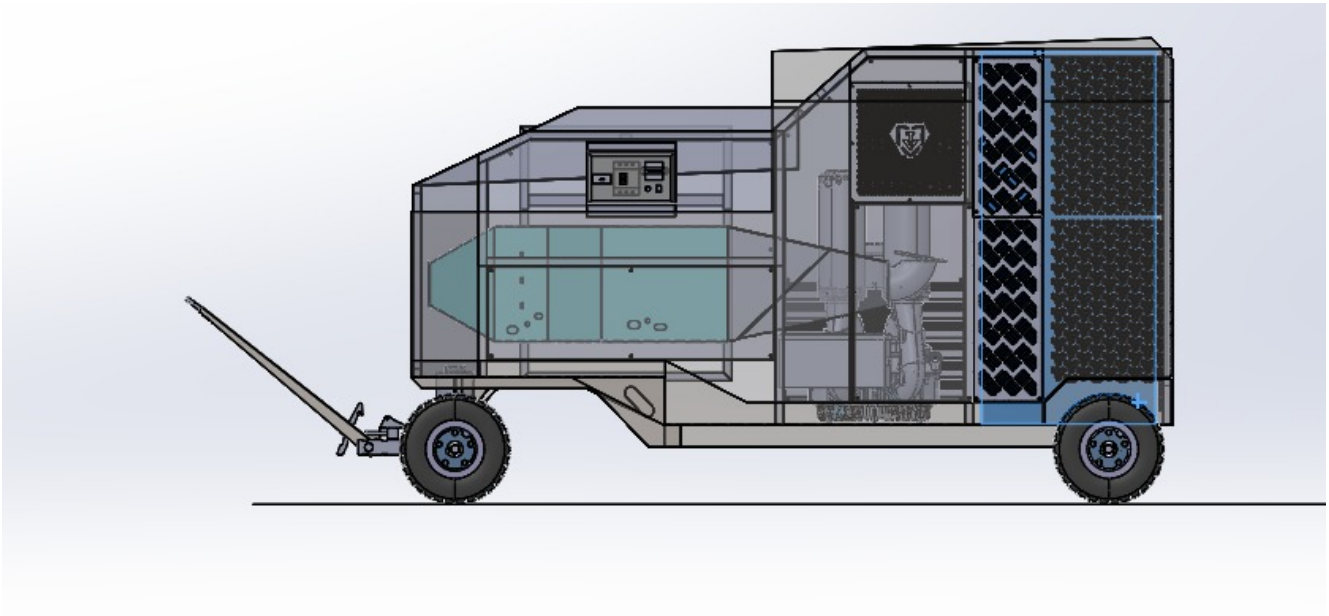
The original CAD screenshots and fabrication photograph are retained together so the design and physical build can be compared directly.

PROJECT 15 / VISUAL RECORD

Dragon AC Cooling Cart



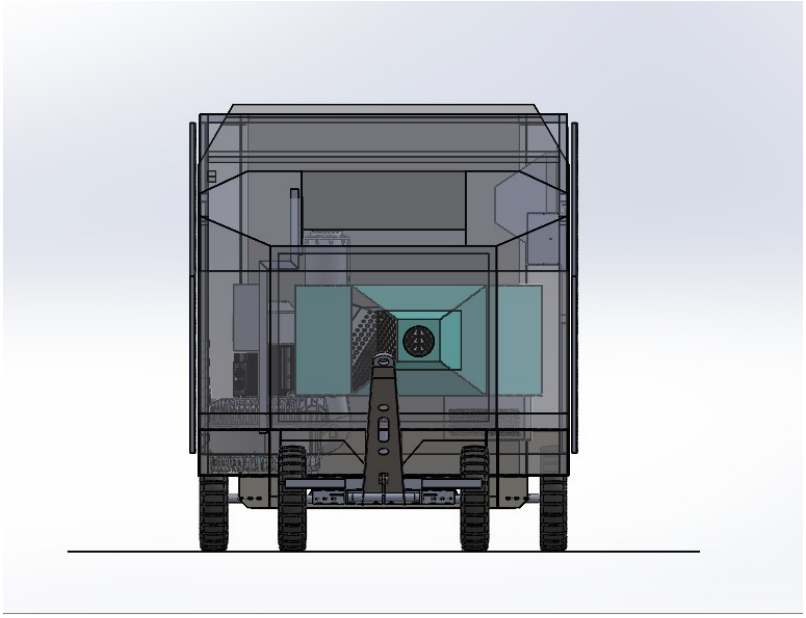
01 / 3D CAD assembly · Isometric view



02 / 3D CAD assembly · Side view

PROJECT 15 / VISUAL RECORD

Dragon AC Cooling Cart



03 / 3D CAD assembly · Front view



04 / Fabricated model · Workshop photograph

PROJECT 16 / ACADEMIC

Academic Layout Design & Equipment Placement

An academic layout study applying Industrial Engineering principles to equipment placement and mounting arrangements. I developed the layout to communicate where components would sit and how their positions relate within the available space.

Layout development

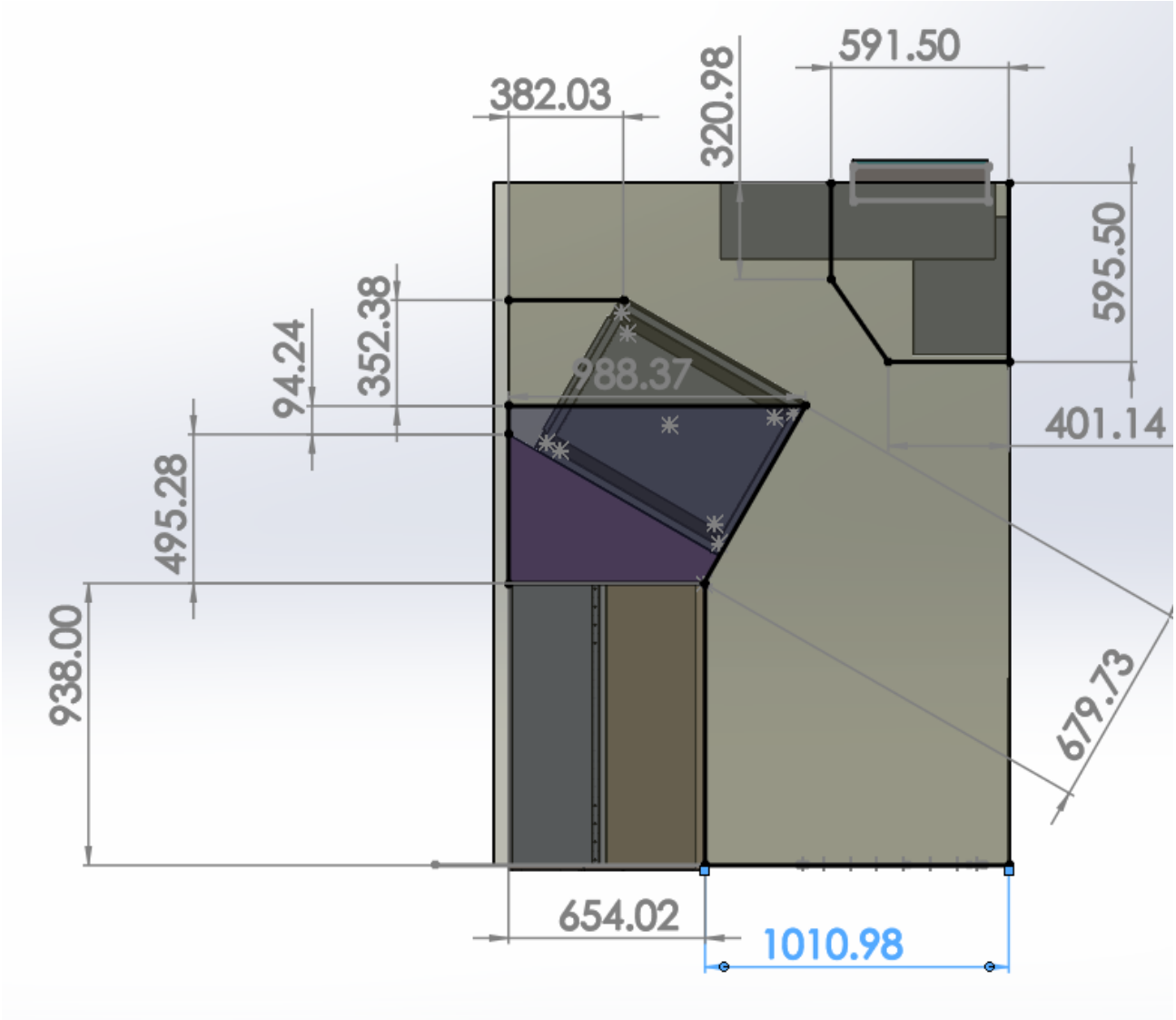
Prepared the CAD layout and placed the individual items within the arrangement, considering their relative positions and proposed mounting locations. The dimensioned view makes these placement decisions easier to communicate and review.

Engineering learning

The work connects academic learning with practical spatial planning and engineering documentation. It demonstrates my approach to organizing equipment clearly before moving into a more detailed design stage.

PROJECT 16 / VISUAL RECORD

Academic Layout Design & Equipment Placement



01 / Dimensioned equipment-placement study

PROJECT 17 / ACADEMIC

Analysis Of Waste And Designing Of Lean Systems for Healthcare

Mapped the complete patient journey from arrival to discharge in a hospital Emergency Department using Value Stream Mapping (VSM). The project applied lean methods to analyze operational waste and improve the flow of care through the department.

Method & contribution

Developed the current-state patient-flow map to understand the full process and identify opportunities for improvement. Applied VSM, 5S and Fishbone analysis to examine workflow issues and support the development of an improved lean system.

Prepared a comparison of the current and improved layouts, making the proposed process changes visible and connecting the analysis to a clearer patient-flow arrangement.

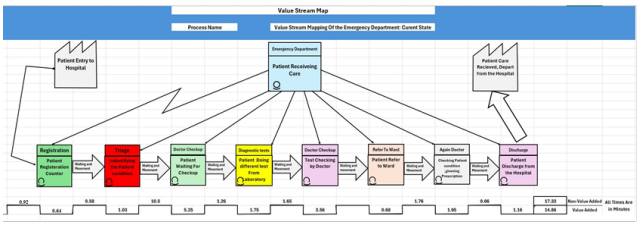
Outcome & academic relevance

The project contributed to reducing service time and operational waste in the Emergency Department through lean improvements to the patient-flow arrangement.

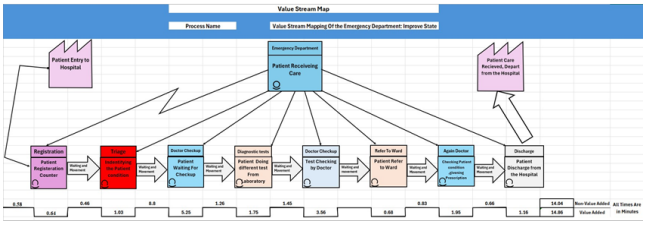
This final year project brings together my interests in Industrial Engineering, systems thinking and process optimization, complementing the mechanical design work presented elsewhere in the portfolio.

PROJECT 17 / VISUAL RECORD

Analysis Of Waste And Designing Of Lean Systems for Healthcare



Current Layout



Improve Layout

01 / Value Stream Mapping · Current and improved layouts

CONTACT

Let’s connect.

Engineer Zeeshan Mumtaz

Industrial Engineer | Design Engineer

Email: qazizeeshan350@gmail.com

WhatsApp: +92 341 3642355

Selected images and drawings document the work presented. Complete CAD models and restricted project details are not distributed through this portfolio.